

MEASUREMENT AND SPACE

AREA AND VOLUME

The largest shopping centre in Australia is Chadstone Shopping Centre in Melbourne. It covers an area of 215 574 square metres, which is about the size of 30 soccer fields. The largest city square in Australia is Adelaide's Victoria Square, measuring 322 m by 161 m, with an area of 51 842 square metres, or 7 soccer fields. This means that about 4.2 Victoria Squares will fit inside Chads one Shopping Centre!



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Chapter outline

	Working mathematically				
8.01 The metric system	U	F			C
8.02 Perimeter	U	F	PS	R	
8.03 Circumference of a circle	U	F	PS	R	
8.04 Metric units for area	U	F		R	C
8.05 Area of a rectangle	U	F	PS	R	
8.06 Area of a triangle	U	F	PS	R	
8.07 Area of a parallelogram	U	F		R	
8.08 Areas of composite shapes	U	F	PS	R	
8.09 Drawing prisms	U	F		R	C
8.10 Metric units for volume	U	F			C
8.11 Volume of a rectangular prism	U	F		R	
8.12 Volume of a prism	U	F		R	
8.13 Volume and capacity	U	F	PS	R	C

Wordbank

area The amount of surface enclosed by a shape

capacity The amount of fluid (liquid or gas) in a container

circumference The perimeter of a circle

perimeter The distance around the outside of a shape

perpendicular height The height of a shape taken at right angles to its base

pi (π) A special irrational number, approximately 3.1416, used in calculating circular measurements

radius The distance from the centre of a circle to the circle's edge

square metre The area of a square that measures 1 m by 1 m

volume The amount of space taken up by a solid

Quiz
Wordbank 8

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (8):

8.01 The metric system
8.02 Perimeter of composite shapes
8.03 Circumference of a circle
8.04 Metric units of area
8.08 Areas of composite shapes
8.10 Metric units of volume
8.12 Volume of a prism
8.13 Volume and capacity

Twig videos (3):

8.01 How long is a metre?
8.09 Escher and the endless staircase • Perspective: Dazzle camouflage

PhET interactive (1):

8.04 Area builder

Quizzes (6):

- Wordbank 8
- SkillCheck 8
- Mental skills 8A
- Mental skills 8B
- Language of maths 8
- Test yourself 8

Skillsheets (4):

SkillCheck Multiplying by 10, 100, 1000
8.04, 8.05 What is area?
8.10 What is volume?
8.12 Solid shapes

Worksheets (19):

8.01 Metric units
8.02 A page of composite shapes
8.03 A page of circles • Circumference and area • Discovering pi
8.04 Areas on a grid • Australian areas
8.05 Australian areas • 1 cm grid paper • Triangle areas
8.06 Rectangle and triangle areas
8.08 A page of composite shapes • Composite areas • Square dot paper
8.09 Square dot paper • Isometric dot paper
8.11 Volume
8.12 Measuring shapes review
8.13 Volume and capacity cards • Volume and capacity • Capacity
Mind map Area and volume

Puzzles (6):

8.01 Converting length units
8.05 Area group clues
8.08 Areas 1 • Area puzzles
8.13 Capacity puzzle
Language of maths Area and volume crossword

Presentation (1):

8.02 Calculating the perimeter

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- use powers of 10 to convert between metric units of length, mass and capacity, and understand the metric prefixes (milli-, centi-, kilo-, mega-) in such units of measurement
- calculate the perimeter of polygons including squares, rectangles and triangles
- identify the radius, diameter and circumference of a circle
- calculate the circumference of a circle
- use and convert between metric units for area and volume
- solve problems involving the areas of rectangles, triangles and parallelograms
- draw different views of prisms
- find the volume of a rectangular prism
- find the volume of a prism
- convert between metric units for volume and capacity, for example, 1 mL is equivalent to 1 cm^3

SkillCheck ANSWERS ON P. 587

1 Copy and complete each conversion.

a	1 kg = _____ g	**b**	1 min _____ s	**c**	1 L = _____ mL
d	1 cm = _____ mm	**e**	1 g = _____ mg	**f**	1 day = _____ hours
g	1 kL = _____ L	**h**	1 t = _____ kg	**i**	1 m = _____ mm
j	1 h = _____ min	**k**	1 m = _____ cm	**l**	1 km = _____ m

2 Evaluate each expression.

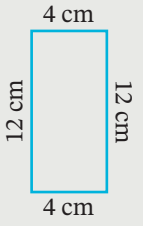
a	10.32×100	**b**	9.45×10	**c**	0.4×1000	**d**	2.7×1000
e	0.9×100	**f**	6.3×10	**g**	$2000 \div 100$	**h**	$43\,000 \div 1000$
i	$650 \div 10$	**j**	$3750 \div 100$	**k**	$95 \div 10$	**l**	$8200 \div 1000$

3 What metric unit would you use to measure:

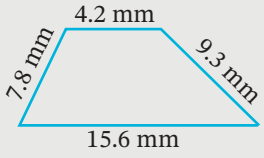
a	the distance from the street to your front door?		
b	the length of your foot?	**c**	the length of an ant?
d	the length of your bed?	**e**	the width of a basketball court?
f	the depth of a river?	**g**	the distance around a football field?
h	the length of a long-jump pit?	**i**	the height of a door?
j	the length of a fingernail?	**k**	an athlete's time in a race?

4 Find the perimeter of each shape.

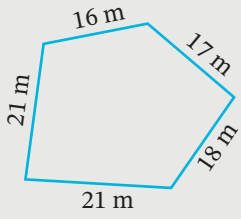
a



b



c



Quiz
SkillCheck 8

Skillsheet
Multiplying by 10, 100, 1000

8.01 The metric system

Video
How long is a metre?

Worksheet
Metric units

Puzzle
Converting length units

In Australia, we measure using the **metric system**. This system began in France in the 1790s after the French revolution and, because it is based on powers of 10, it is logical and easy to use. The word 'metric' comes from the Greek word *metron* meaning 'to measure'.

In 1970, the Metric Conversion Board was established in Australia, starting the change to metric units, shown in the table.



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Quantity	Name of unit	Abbreviation
Length	metre	m
	millimetre	mm
	centimetre	cm
	kilometre	km
Mass	kilogram	kg
	gram	g
	milligram	mg
	tonne	t
Time	second	s
	minute	min
	hour	h
	day	d
Area	square metre	m^2
	square centimetre	cm^2
	hectare	ha
	square kilometre	km^2
Volume	cubic metre	m^3
	cubic centimetre	cm^3
Capacity	litre	L
	millilitre	mL
	kilolitre	kL
	megalitre	ML
Temperature	degree Celsius	°C

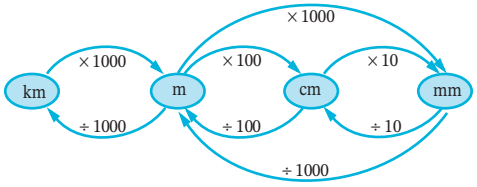
The metric system uses the following **prefixes** for metric units.

Prefi	Abbreviation	Meaning	Example
micro-	μ	one-millionth: $\frac{1}{1\,000\,000}$	1 μm = $\frac{1}{1\,000\,000}$ m
milli-	m	one-thousandth: $\frac{1}{1000}$	1 mg = $\frac{1}{1000}$ g
centi-	c	one hundredth: $\frac{1}{100}$	1 cm = $\frac{1}{100}$ m
kilo-	k	one thousand times: 1000 ×	1 km = 1000 m
mega-	M	one million times: 1 000 000 ×	1 ML = 1000 kL

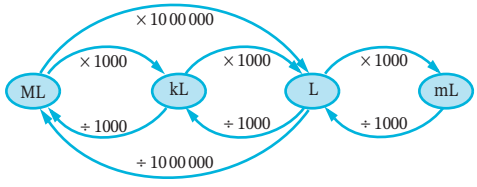
9780170477963

This table shows the relationships between commonly-used units of measurement.

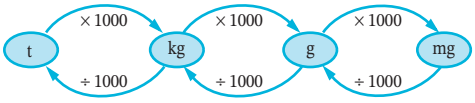
Length 1 cm = 10 mm
1 m = 100 cm = 1000 mm
1 km = 1000 m



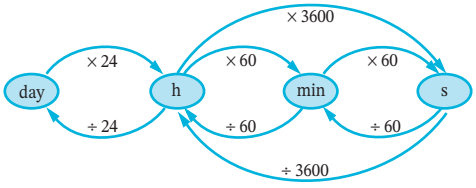
Capacity 1 L = 1000 mL
1 kL = 1000 L
1 ML = 1000 kL = 1 000 000 L



Mass 1 g = 1000 mg
1 kg = 1000 g
1 t = 1000 kg



Time 1 min = 60 s
1 h = 60 min = 3600 s
1 day = 24 h



Example 1

Convert:

a 13 km to m
b 5.8 t to kg
c 16 000 mL to L
d 1500 min to h

Solution

a 13 km = 13 × 1000 m
= 13 000 m

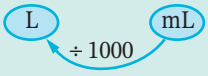


b 5.8 t = 5.8 × 1000 kg
= 5800 kg

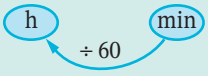


To convert large to small units, multiply (×).

c 16 000 mL = 16 000 ÷ 1000 L
= 16 L



d 1500 min = 1500 ÷ 60 h
= 25 h



To convert small to large units, divide (÷).

Video
The metric system

EXERCISE 8.01 ANSWERS ON P. 588

The metric system UFC

EXAMPLE 1

1 Copy and complete:

a 1 h = _____ min
b 1 km = _____ m
c 1 kg = _____ g
d 1 cm = _____ mm
e 1 day = _____ h
f 1 kL = _____ L

Foundation Standard Complex

g 1 t = _____ kg
h 1 min = _____ s
i 1 L = _____ mL
j 1 g = _____ mg
k 1 ML = _____ L
l 1 h = _____ s

2 Convert:

a 6.4 m to cm
b 4 t to kg
c 3 days to hours
d 2.4 ML to L
e 0.9 m to mm
f 12.7 kg to g
g 8 h to min
h 5.6 L to mL
i 42 cm to mm
j 72 g to mg
k 3.8 km to cm
l 4.5 h to s

3 Convert each length to centimetres.

a 2 m
b 1500 mm
c 0.5 m
d 3.5 m
e 0.25 m
f 2 000 000 μm
g 15.8 km
h 0.4 km

4 Convert each length to metres.

a 200 cm
b 3000 mm
c 850 cm
d 9800 mm
e 75 cm
f 325 mm
g 6 km
h 18 km

5 Write the metric prefix that means: C

a 1000
b $\frac{1}{100}$
c $\frac{1}{1000}$

6 Find other prefixes that are used in the metric system. Write their meanings. C

7 Copy and complete each conversion.

a 4.5 kg =_____ g
b 3 hours =_____ minutes
c 750 mL =_____ L
d 240 mm =_____ cm
e 12 kL =_____ L
f 5.2 m =_____ cm
g 420 s =_____ min
h 2 days =_____ hours
i 8.65 t = _____ kg
j 72 hours =_____ days
k 65 mm = _____ m
l 0.96 kL = _____ L

8 Which length is the shortest? Select the correct answer **A**, **B**, **C** or **D**.

A 0.3 m
B 29 cm
C 300 mm
D 160 mm

9 Karla and Cassie measured these objects in the classroom. C

- length of pencil 12 cm
- width of computer screen 33 cm
- width of door 80 cm
- width of ruler 3.5 cm
- length of room 740 cm

a Karla wrote her measurements in millimetres. What did she write?

b Cassie wrote her measurements in metres. What did she write?

c What unit would you use (cm, mm or m)? Would you use different units for different objects? If so, what would they be?

10 Shanina is in Year 7. Select the most reasonable answer for Shanina's height in centimetres.

A 14 **B** 50 **C** 150 **D** 1200

11 Copy and complete each conversion.

a 2.2 L = _____ mL **b** 800 mm = _____ m

c 235 cm = _____ m **d** 360 mg = _____ g

e $5\frac{1}{2}$ hours = _____ min **f** 4900 L = _____ kL

g 270 kg = _____ t **h** 4 years = _____ months

i 10.8 km = _____ m **j** 16.9 L = _____ mL

k 1 year = _____ days **l** 740 cm = _____ m

12 Shahid wishes to fill a 3-litre bucket using a 250-millilitre cup. How many full cups will he need? Select **A**, **B**, **C** or **D**.

A 12 **B** 83 **C** 750 **D** 750 000

Perimeter 8.02

Perimeter

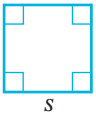
The **perimeter** of a shape is the distance around the shape.

It is the sum of the lengths of the sides of the shape.

Worksheet
A page of composite shapes

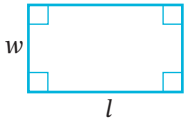
Video
Perimeter of composite shapes

The **perimeter of a square** can be found by adding all of the sides:



$$\text{Perimeter} = s + s + s + s$$
$$= 4s$$

The **perimeter of a rectangle** can be found by adding all the sides:



$$\text{Perimeter} = l + w + l + w$$
$$= 2l + 2w \text{ or } 2(l + w)$$

Perimeter of a square and rectangle

Perimeter of a square

$P = 4s$, where s is the length of one side.

Perimeter of a rectangle

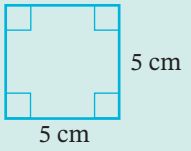
$P = 2l + 2w$ or $2(l + w)$, where l is the length and w is the width (breadth).

Foundation Standard Complex

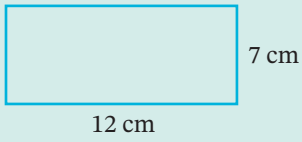
Example 2

Find the perimeter of each shape.

a



b



Solution

a Perimeter of a square $= 4s$

$= 4 \times 5$

$= 20$ cm

b Perimeter of a rectangle $= 2(l + w)$

$= 2 \times (12 + 7)$

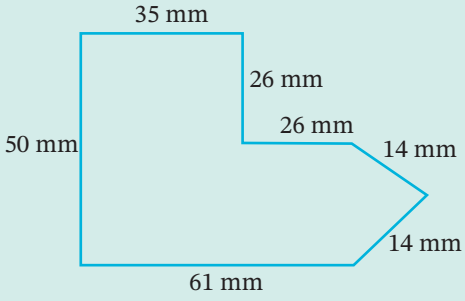
$= 38$ cm

Presentation
Calculating the perimeter

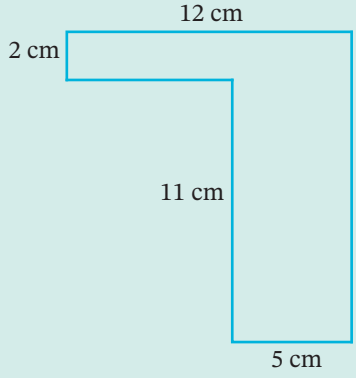
Example 3

Find the perimeter of each shape.

a



b

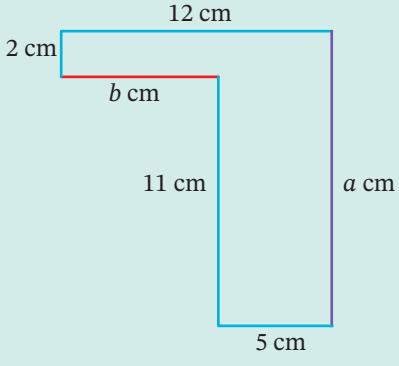


Solution

a Perimeter $= 50 + 35 + 26 + 26 + 14 + 14 + 61$ Adding side lengths

$= 226$ mm

b First, we need to find the lengths of the unknown sides, a and b.



$a = 2 + 11 = 13$ cm The side shown by the purple line

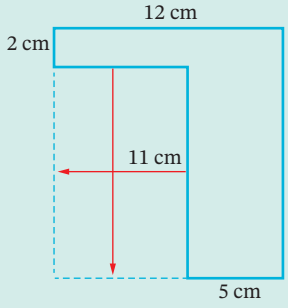
$b = 12 - 5 = 7$ cm — The side shown by the red line

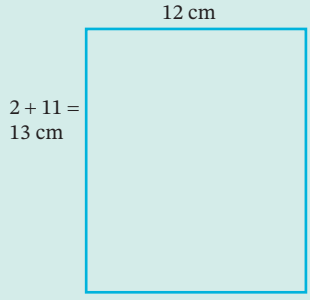
Perimeter $= 12 + 13 + 5 + 11 + 7 + 2$

$= 50$ cm

Alternative solution

Two sides of any L-shape can be moved to make a rectangle, whose perimeter is easier to find.





Perimeter $= 2(l + w)$

$= 2(13 + 12)$

$= 50$ cm

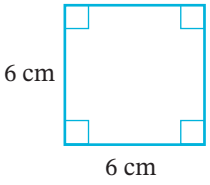
EXERCISE 8.02 ANSWERS ON P. 588

Perimeter U F PS R

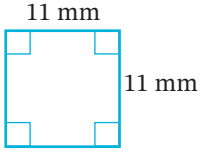
1 Find the perimeter of each shape.

EXAMPLE 2

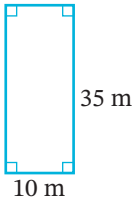
a



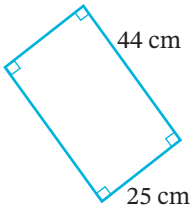
b



c



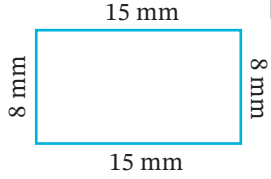
d



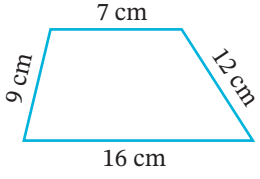
2 Find the perimeter of each shape.

EXAMPLE 3

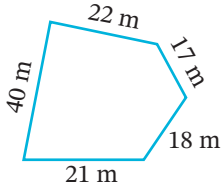
a



b

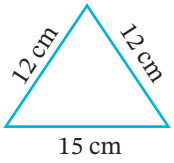


c

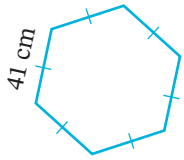


Foundation | Standard | Complex

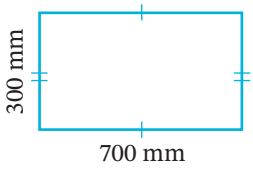
d



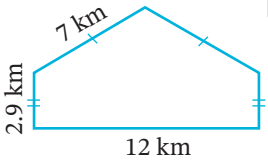
e



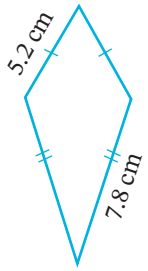
f



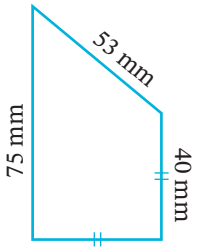
g



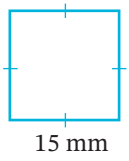
h



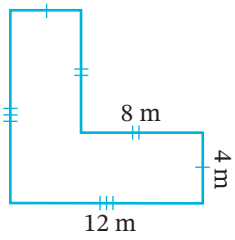
i



j

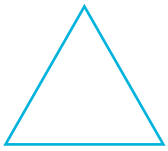


k

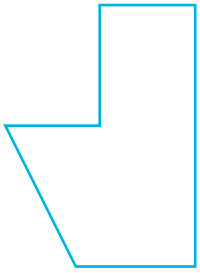


3 Measure the sides of each shape in mm and find its perimeter.

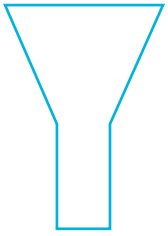
a



b



c

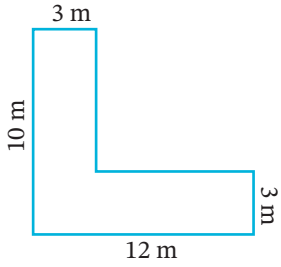


d

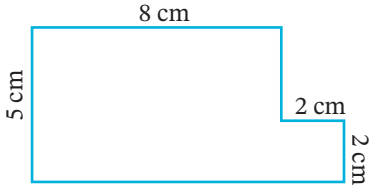


4 For each figure, find the missing lengths and then its perimeter. R

a



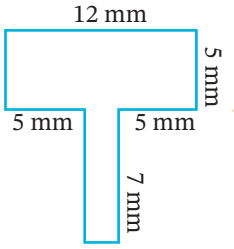
b



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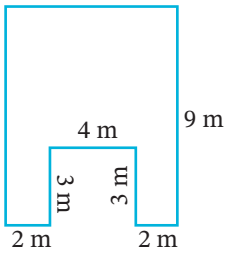
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c

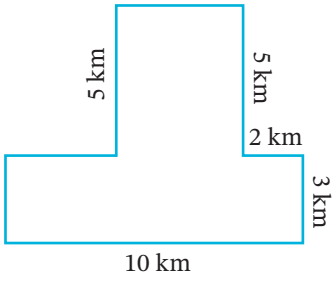


Find a better way of calculating the perimeter of a T-shape.

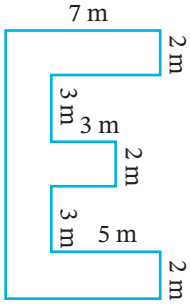
d



e



f

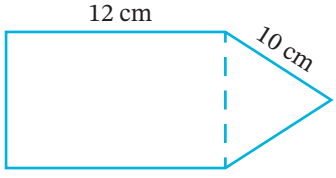


5 A rectangle has a perimeter of 64 m. What could its length and width be (in metres)? Select the correct answer **A**, **B**, **C** or **D**. R

A 32, 32 **B** 49, 15 **C** 16, 18 **D** 18, 14

6 A rectangle has a perimeter of 30 cm. What could its length and width be? R

7 This figure is made up of a rectangle and an equilateral triangle. Find its perimeter. Select **A**, **B**, **C** or **D**.



A 44 cm **B** 64 cm

C 54 cm **D** 22 cm

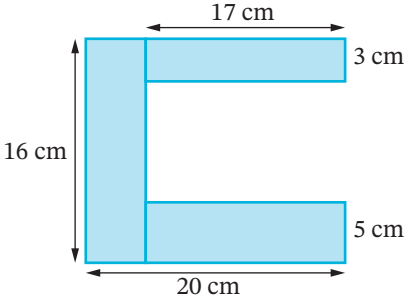
8 **a** A square has a perimeter of 16 cm. Find the length of one side.

b A rectangle has a perimeter of 40 cm. If its length is 12 cm, find its width.

c A square has a perimeter of 51 cm. Find the length of one side.

d A rectangle has a perimeter of 62 cm. If its width is 13 cm, find its length. R

9 Dana built the letter C using 3 planks of wood as shown in the figure. PS R



Find the perimeter of the wooden letter in centimetres. Select **A**, **B**, **C** or **D**.

A 61 **B** 72 **C** 122 **D** 106

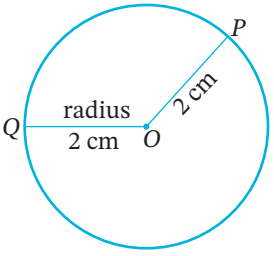
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8.03 Circumference of a circle

Worksheets
A page of circles

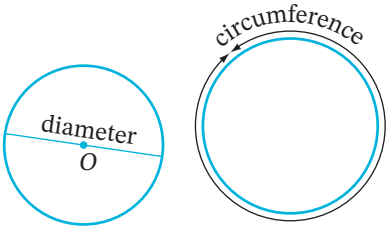
Circumference and area

The circle is a completely round shape. Every point on a circle is the same distance from its centre. The circle shown has centre O, and points P and Q are exactly 2 cm from O. This distance, from the centre to the edge of the circle, is called the **radius** of the circle (*plural*: radii).



An interval running from one side of a circle to the other side and through the circle's centre is called the **diameter** of the circle. The length of a circle's diameter is **double** the length of its radius.

The perimeter of a circle is called the **circumference**.



Investigation

Measuring the circumference of a circle

Worksheet
Discovering pi

This is a measuring activity to be done in groups of 3.

You will need: A measuring tape, or some string and a ruler; graph paper; 6 round objects such as cans, round cake tins, pipes, coins, drums, bottles, and so on.

1 Copy this table.

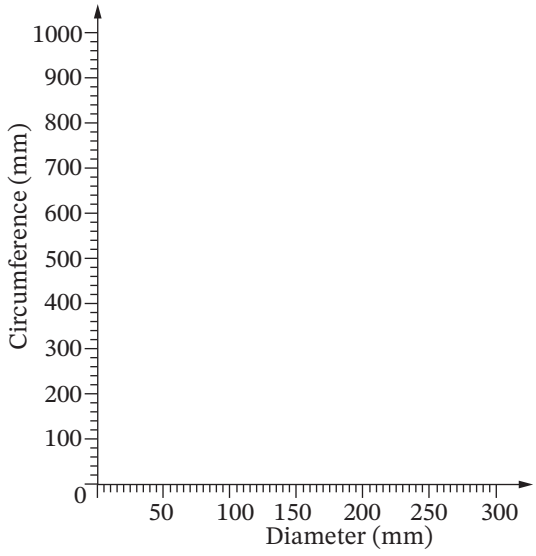
Object	Diameter, d (mm)	Circumference, C (mm)	The ratio $\frac{C}{d}$
a			
b			
c			
d			
e			
f			

2 Measure (in millimetres) to find the diameter and the circumference of 6 different objects. Record your results in the table.

3 Calculate the ratio $\frac{C}{d}$ for each object. Round your answers to 2 decimal places.

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4 Copy and complete the graph, using your results from the table. What do you notice about the graph?



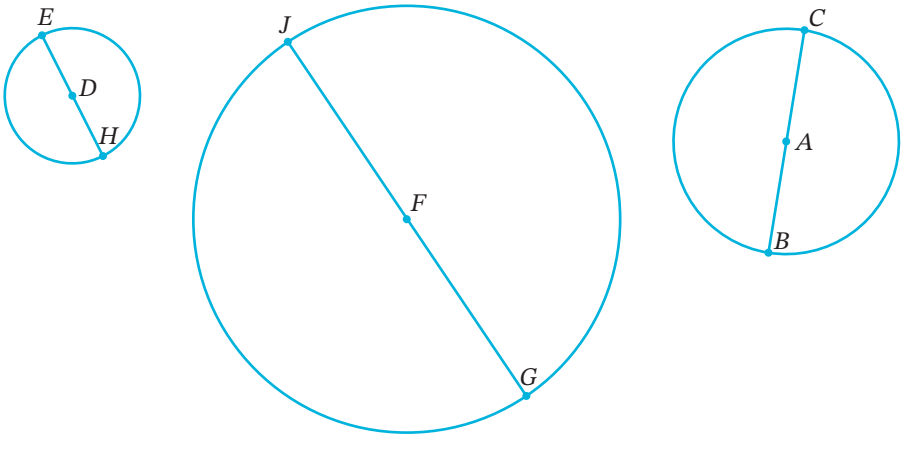
5 There is a formula for finding the circumference of a circle.

Copy and complete the formula: $C = d \times$ _____

Technology

The circumference and diameter of a circle

1 Use your dynamic geometry software to construct 3 different circles. Construct the diameter of each circle.



2 Calculate each circle's circumference and diameter.

3 For each circle, use your calculator to find $\frac{\text{length of circumference}}{\text{length of diameter}}$

4 What do you notice? Discuss this with another student.

The **circumference** of a circle can be found by multiplying its **diameter** by a special number called **pi** (pronounced 'pie'), represented by the Greek letter π. For any circle, $\frac{\text{circumference}}{\text{diameter}} = \pi = 3.14\ldots$

Pi is often estimated as 3.14, but a more accurate value can be found on your calculator when you press the π key (you may need to press SHIFT first). As a decimal, the digits of π continue endlessly without any repeats or patterns:

$\pi = 3.141\ 592\ 653\ 589\ 793\ \ldots$

so, like $\sqrt{2}$, it is called an **irrational number**.

This number was named 'pi' by the Swiss mathematician Leonhard Euler in 1737.

The formula for the circumference of a circle is $C = \pi \times \text{diameter} = \pi d$.

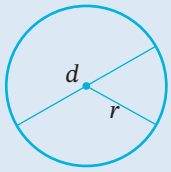
Because the diameter of a circle is double its radius, the circumference, C, of a circle with radius r is $C = \pi \times 2 \times \text{radius} = 2\pi r$.

Circumference of a circle

The **circumference (perimeter) of a circle** is:

$C = \pi \times \text{diameter}$ or $C = 2 \times \pi \times \text{radius}$

$C = \pi d$ or $C = 2\pi r$



Video
Circumference of a circle

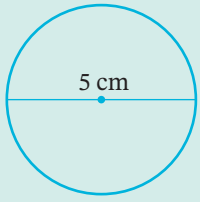
Example 4

a Estimate the circumference of a circle with a diameter of 5 cm.

b Calculate the circumference of the circle:

i correct to 2 decimal places

ii in terms of π



Solution

a $C = \pi d$

$= \pi \times 5$

$\approx 3 \times 5$ π is approximately 3.

$= 15$ cm

b **i** $C = \pi d$ On your calculator enter π $\times$ 5 $=$.

$= \pi \times 5$

$= 15.707\ 963\ldots$

≈ 15.71cm

ii $C = \pi d$

$= \pi \times 5$

$= 5\pi$ cm

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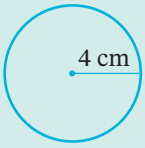
Like writing an answer as a surd in Pythagoras' theorem problems, writing an answer in terms of π is more exact as there is no rounding involved.

Example 5

Calculate the circumference of the circle:

a correct to one decimal place

b in terms of π



Solution

a $C = 2\pi r$
$= 2 \times \pi \times 4$
$= 25.13274\ldots$
≈ 25.1 cm

b $C = 2\pi r$
$= 2 \times \pi \times 4$
$= 8\pi$ cm

Video
Circumference of a circle

8.03

EXERCISE 8.03 ANSWERS ON P. 588

Circumference of a circle U F PS R

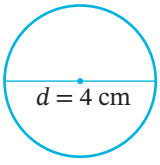
1 Use a ruler and compasses to construct a circle of:

a radius 4 cm **b** radius 2.5 cm **c** radius 56 mm

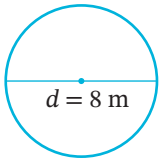
EXAMPLE 4

2 Estimate the circumference of each circle.

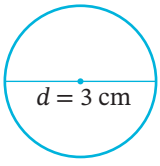
a



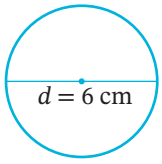
b



c



d



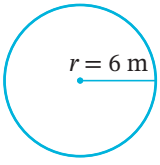
3 Calculate, correct to 2 decimal places, the circumference of each circle in question **1**.

4 Calculate the circumference of each circle in question **1** in terms of π.

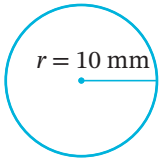
EXAMPLE 5

5 Estimate the circumference of each circle.

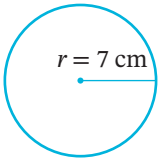
a



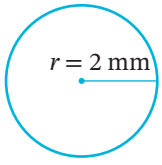
b



c



d



6 Calculate, correct to one decimal place, the circumference of each circle in question **4**.

7 Calculate the circumference of each circle in question **4** in terms of π.

8 A child's inflatable swimming pool has a diameter of 1.4 m. Find its circumference correct to 2 decimal places.

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9 Tina's bicycle has wheels with a diameter of 60 cm. PS

a How far correct to one decimal place does the bicycle move when a wheel turns around once?

b If Tina cycles 900 m to school, how many complete turns does the bicycle wheel make?

10 The Earth has a radius of 6370 km. Find correct to one decimal place the distance around the Equator.

11 A 20-cent coin has a radius of 14 mm. Calculate its circumference correct to one decimal place.

12 This tin of tomatoes has a diameter of 75 mm. If the label wraps around the tin completely, how long is the label? Answer correct to the nearest millimetre. PS R



Shutterstock.com/Bahadir Yeniceri

13 Calculate, correct to 2 decimal places, the circumference of a circle with:

a a diameter of 2 cm

b a diameter of 7 cm

c a radius of 3 cm

d a radius of 5 cm

14 Which of the following intervals best shows the circumference of this circle? Select **A**, **B**, **C** or **D**. R



A _____

B _____

C _____

D _____

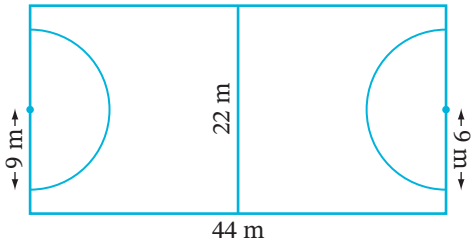
15 The International Space Station completes one circular orbit of the Earth at a height of 400 km. If the radius of the Earth is approximately 6400 km, find the distance travelled by the station. Select the closest answer **A**, **B**, **C** or **D**.

A 2513 km **B** 37 699 km **C** 40 212 km **D** 42 726 km

16 Tape has been placed on all the lines of this indoor hockey pitch. How much tape was used? Express the answer: PS R

a correct to one decimal place

b in terms of π.

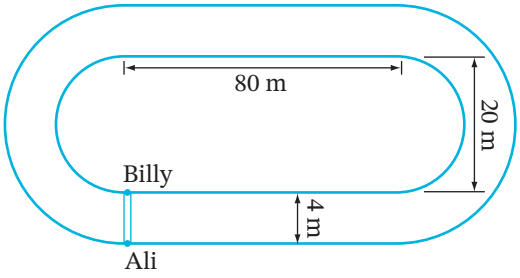


17 A circle has a circumference of 50.265 cm. Find its diameter, correct to the nearest centimetre. R

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18 Ali and Billy raced each other around this athletic track. Ali ran along the outside perimeter while Billy ran along the inside perimeter. After one lap of the track, who ran the longer distance, and by how much? Answer correct to the nearest metre. PS R



Did you know?

Never-ending pi!

$\pi = 3.141\ 592\ 653\ 589\ 793\ 238\ 462\ 643\ 383\ 279\ 502\ 884\ 197\ 169\ 399\ 375\ 105\ 820...$

Ancient civilisations knew about the value of π, estimating it as 3. Over time, the calculations have improved due to better formulas and technology. Since the first computer, the ENIAC, was invented in 1949, much progress has been made. In 2019 Emma Haruka Iwao, a Google employee from Japan, calculated the value of π to 31.4 trillion digits with the help of Google's cloud computing service.

Year	Person/Country	Number of decimal places
1855	Shanks, England	527
1949	ENIAC computer, USA	2037
1973	CDC 7600 computer, France	1 000 000
1988	Kanada, Hitachi S-820 computer, Japan	200 000 000
1989	Chudnovsky brothers, USA	1 000 000 000
1999	Kanada and Takahashi, Hitachi SR8000 computer, Japan	206 158 430 000
2002	Kanada, Hitachi SR8000 computer, Japan	1 241 100 000 000
2010	Shigeru Kondo, Alexander Yee, Japan and USA	5 trillion
2019	Emma Haruka Iwao, Google Cloud, USA	31.4 trillion
2022	Emma Haruka Iwao, Google Cloud, USA	100 trillion

The Guinness World Record for reciting the most digits of pi belongs to Rajveer Meena of India, who recited pi to 70 000 decimal places (while blindfolded) in 2015. Unofficially, in 2006, Japanese counsellor Akira Haraguchi memorised π to 100 000 decimal places. It took him over 16 hours to recite it. He had to stop after 3 hours as he lost his place, and had to start from the beginning.

On average, how many digits would Haraguchi have recited per minute?

8.04 Metric units for area

Skillsheet
What is area?

Worksheets
Areas on a grid
Australian areas

Interactive
Area builder

Video
Metric units of area

A person estimating the number of tiles needed to cover the floor of an outdoor space needs to know the size of a tile and the size of the floor. In this case, the **area** of the floor gives its 'size'.

Area

The **area** of a shape is the amount of surface that is enclosed by the shape.

For example, the area of this trapezium has been shaded.



The table shows metric units for area.

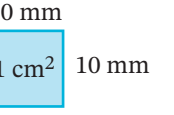
Area unit	Abbreviation	The size of a square with each side measuring
square millimetre	mm^2	1 mm (Actual size: ▪)
square centimetre	cm^2	1 cm (Actual size: ▪)
square metre	m^2	1 m
hectare	ha	100 m

A **square centimetre** is about the area of a fingernail.

$1 \text{ cm} = 10 \text{ mm}$

$1 \text{ cm}^2 = 10 \times 10 \text{ mm}^2$

$= 100 \text{ mm}^2$ (double the number of 0s)

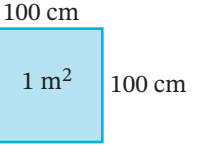


A **square metre** is approximately the size of the floor of a large shower recess.

$1 \text{ m} = 100 \text{ cm}$

$1 \text{ m}^2 = 100 \times 100 \text{ cm}^2$

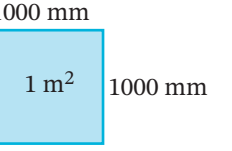
$= 10\,000 \text{ cm}^2$ (double the number of 0s)



$1 \text{ m} = 1000 \text{ mm}$

$1 \text{ m}^2 = 1000 \times 1000 \text{ mm}$

$= 1\,000\,000 \text{ mm}^2$ (double the number of 0s)

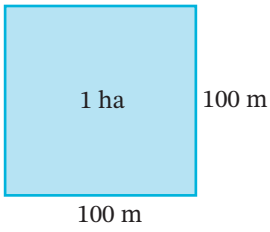


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A **hectare** is about the area of Sydney Football Stadium, or the grassed area inside an athletics (400 m) track.

$1 \text{ ha} = 100 \times 100 \text{ m}$

$= 10\,000 \text{ m}^2$ (double the number of 0s)



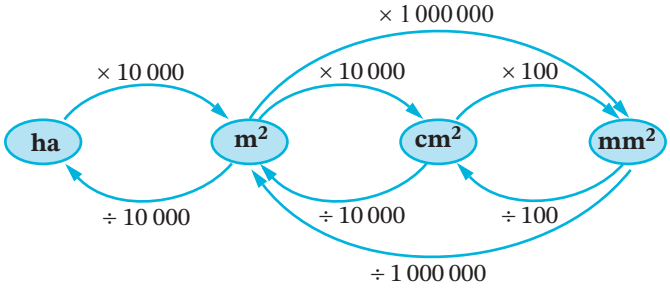
Metric units for area

$1 \text{ cm}^2 = 100 \text{ mm}^2$

$1 \text{ m}^2 = 10\,000 \text{ cm}^2 = 1\,000\,000 \text{ mm}^2$

$1 \text{ ha} = 10\,000 \text{ m}^2$

This diagram shows how to convert between different units of area.



Example 6

Convert

a 3 cm^2 to mm^2 **b** 4000 mm^2 to m^2 **c** $81\,000 \text{ m}^2$ to ha

Solution

a $3 \text{ cm}^2 = 3 \times 100 \text{ mm}^2$ cm^2 to mm^2 : $\times 100$

$= 300 \text{ mm}^2$

b $4000 \text{ mm}^2 = 4000 \div 1\,000\,000 \text{ m}^2$ mm^2 to m^2 : $\div 1\,000\,000$

$= 0.004 \text{ m}^2$

c $81\,000 \text{ m}^2 = 81\,000 \div 10\,000 \text{ ha}$ m^2 to ha : $\div 10\,000$

$= 8.1 \text{ ha}$

EXERCISE 8.04 ANSWERS ON P. 588

Metric units for area U F R C

1 What unit of area would you use to measure the area of: R C

a a farm? **b** a classroom? **c** a shirt?

d the school oval? **e** a sheet of paper? **f** your eardrum?

2 What is the approximate area of a calculator? Select the correct answer a **A**, **B**, **C** or **D**. R C

A 80 cm^2 **B** 120 cm^2 **C** 160 cm^2 **D** 200 cm^2

EXAMPLE 6

3 Copy and complete each conversion.

a 15 cm^2 = _____ mm^2 **b** 1 500 000 mm^2 = _____ m^2

c 690 mm^2 = _____ cm^2 **d** 6.5 m^2 = _____ cm^2

e 0.5 m^2 = _____ mm^2 **f** 12 ha = _____ m^2

g 12 200 cm^2 = _____ m^2 **h** 1250 mm^2 = _____ cm^2

i 0.32 ha = _____ m^2 **j** 7.9 cm^2 = _____ mm^2

k 0.75 m^2 = _____ mm^2 **l** 2 450 000 m^2 = _____ ha

m 865 000 cm^2 = _____ m^2 **n** 51 300 mm^2 = _____ m^2

o 0.47 m^2 = _____ cm^2 **p** 9000 m^2 = _____ ha

4 A square tile has an area of 1024 cm^2. Write this area in:

a mm^2 **b** m^2

5 **a** Arrange 6.5 m^2, 114 000 cm^2 and 25 050 000 mm^2 in ascending order.

b Arrange 990 mm^2, 54 cm^2, 0.000 032 m^2 in descending order. R C

6 Sydney's CBD (central business district) is approximately a rectangle bounded by George Street, Circular Quay, Macquarie and College streets and Liverpool Street, an area of 875 000 square metres. What is this area:

a in hectares? **b** in square centimetres?

7 What is the approximate area of a 5-cent coin? Select **A**, **B**, **C** or **D**.

A 25 mm^2 **B** 2.5 cm^2 **C** 25 cm^2 **D** 2.5 m^2

8 Western Australia is Australia's largest state, with an area of 2 529 875 km^2. Tasmania is our smallest state, with an area of 68 401 km^2.

a What is the area of Tasmania in hectares?

b How many 'Tasmanias' would fit into Western Australia?

9 A postage stamp has an area of 8.36 cm^2. What is this area in square millimetres?

10 The theme park, Dreamworld, on Queensland's Gold Coast, has an area of one square kilometre. R C

a How many square metres are there in a square kilometre?

b How many hectares are there in a square kilometre?

c Name something that might be measured in square kilometres.

Did you know?

The Roman Colosseum

The Colosseum is a famous and impressive ancient building in Rome, Italy, the greatest amphitheatre of the Roman empire. It was built in 80 CE and you can still visit its ruins today. This was the place where huge crowds of Romans watched gladiator fights, animal hunts, executions and other blood-thirsty events.



Shutterstock.com/Fabio Mancino Photography

The Colosseum is oval-shaped, 189 metres long, 16 metres wide and covers an area of about 24 000 square metres. It was surrounded by a wall 48 metres high and could hold over 50 000 spectators.

Find out how the size of the Colosseum compares with the Sydney Cricket Ground or Melbourne Cricket Ground.

Investigation

Standing room only!

1. As a group, mark out a square metre in your classroom or school yard.
2. Estimate how many students could stand in the square.
3. Check your estimate by getting as many students as possible to stand in the square.
4. How could you estimate whether all the Year 7 students in your school could stand in your classroom?
5. How many Year 7 students could stand in your room?

Investigation

The human lagoon

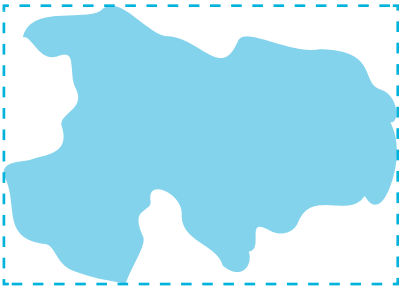
1 **a** Find a level area of concrete or bitumen. Carefully pour a litre of water over it to form a puddle.

b Use a metre ruler or tape measure to measure the dimensions of the smallest rectangle that surrounds the puddle.

c On 1 cm grid paper, make a scale drawing of the rectangle that surrounds the puddle.

d Sketch the shape of your puddle inside the rectangle on the graph paper.

e Use the method of counting squares to find the area of the puddle formed by the litre of water.



2 **a** About 80% of the human body is made up of water. If 1 litre of water has a mass of 1 kilogram, measure your own mass and use it to calculate the number of litres of water in your body. Use this result, together with those from question **1**, to find the area of the puddle the water in your body would make.

b Go outside and mark out a rectangle that has the same area as the puddle you would make. Record the dimensions of this rectangle.

c Using the scale 1 cm = 1 m, draw a rectangle on graph paper that has the same area as the puddle you would make. Write a brief report of what this rectangle represents and how it was found.

Investigation

Student space in a classroom

1 As a group, shift all the furniture to the sides of the classroom.

2 Use a set of metre-long sticks and connectors to lay out a grid of square metres on the floor.

3 Calculate or count the area of the floor in metres.

4 Sit 2 people to a square metre. Are you comfortable enough to stay within your square for a whole class?

5 Sit one person to a square metre. How comfortable are you?

6 Replace the furniture over the grid and sit at your desk.

7 How much area do you and your desk occupy?

8 During the COVID-19 pandemic in 2020, social distancing measures were introduced to minimise the spread of the virus. If each student was allocated an area of 4 square metres, how many students could fit inside your classroom?

9 Copy and complete each sentence.

a My class has _____ students.

b Today my maths class was held in a room with floor dimensions _____ by _____.

c The area of the floor is approximately _____.

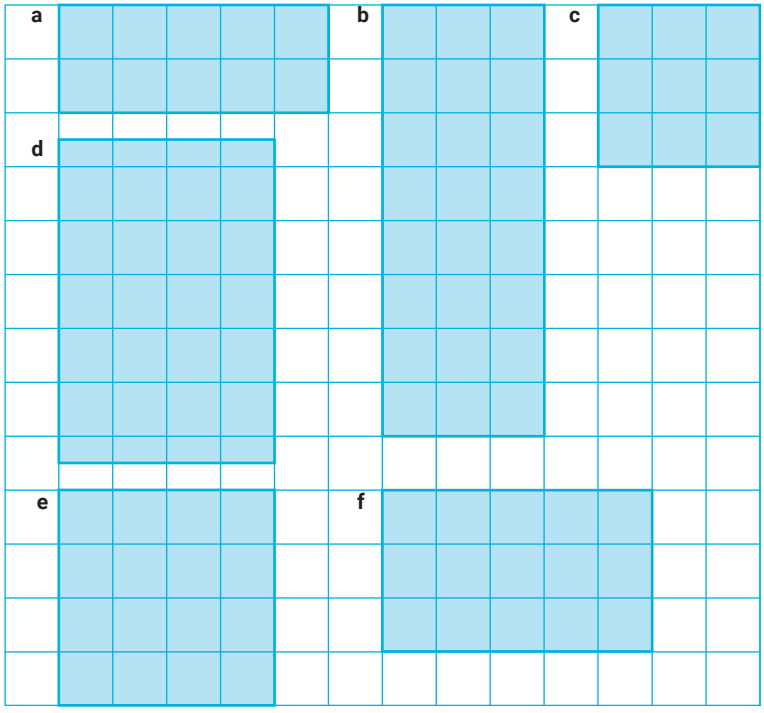
d Each student is allocated _____ of floor space.

e The floor space left over for aisles and the teacher is _____.

Investigation

Area of a rectangle

1 The rectangles and squares below are drawn on a grid of 1 cm squares. Find the area of each rectangle and square (in square centimetres) by counting the squares in it. Copy and complete the table. For rectangles, we usually call the *shorter* measurement the width.



Rectangle	Length (cm)	Width (cm)	Area (cm^2)
a	5	2	
b	8		
c			
d			
e			
f			

2 Look at your results for question **1**. Write down a rule for finding the area of a rectangle or square from its length and width.

Worksheet
Areas on a grid

8.05 Area of a rectangle

Skillsheet
What is area?

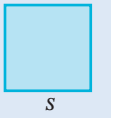
Worksheet
Australian areas

Puzzle
Area group clues

Area of a square

Area = side × side = $s \times s$

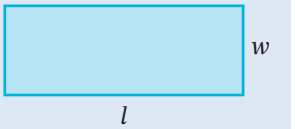
$A = s^2$



Area of a rectangle

Area = length × width

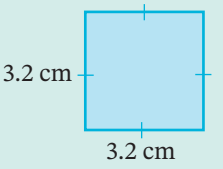
$A = lw$



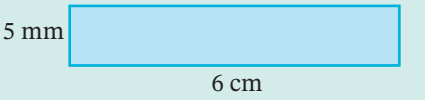
Example 7

Find the area of each shape.

a



b



Solution

a Area = s^2

$= 3.2^2$

$= 3.2 \times 3.2$

$= 10.24 \text{ cm}^2$

b Area = $l \times w$

$= 60 \times 5$

$= 300 \text{ mm}^2$

Change 6 cm to 60 mm to make units the same

It is also possible to change 5 mm to 0.5 cm to have an answer of 3 cm^2 instead.

EXERCISE 8.05 ANSWERS ON P. 588

Area of a rectangle U F PS R

EXAMPLE 7

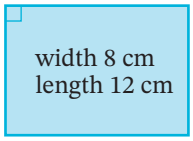
1 Find the area of each rectangle.

a length 9.3 m, width 6 m

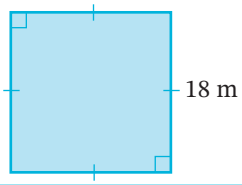
b length 29.3 m, width 2 m

9780170477963

c



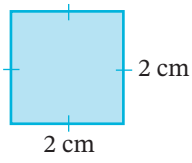
d



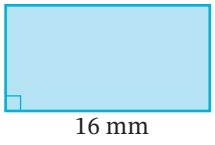
e



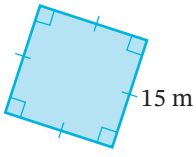
f



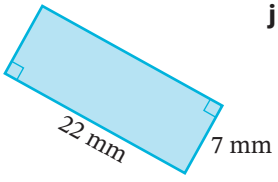
g



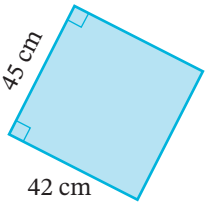
h



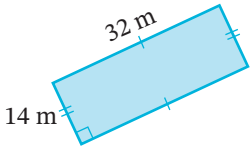
i



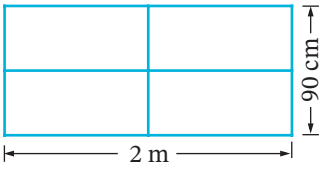
j



k



2 Find the area of this window in square metres.
Select the correct answer **A**, **B**, **C** or **D**.



A 0.18 m²
B 1.8 m²
C 18 m²
D 180 m²

3 Find the area of each item described.

i in m² **ii** in cm² **iii** in mm²

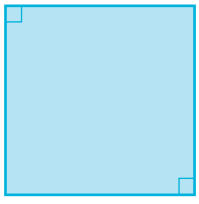
a A square bowling green of length 5 m

b A desk top measuring 95 cm by 54 cm.

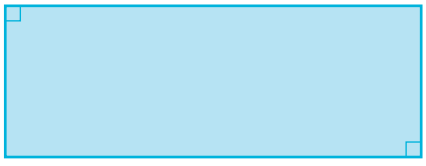
4 Measure the sides of each rectangle and calculate its area:

i in mm² **ii** in cm²

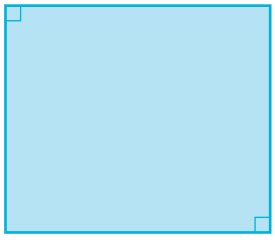
a



b



c



d



Foundation | Standard | Complex

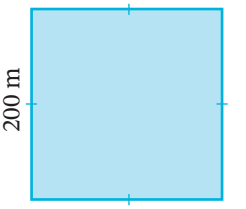
5 A square has an area of 49 cm^2. What is the length of one side?

6 Find the area of each rectangle in square metres.

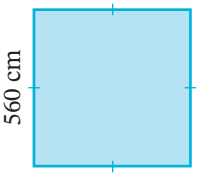
a



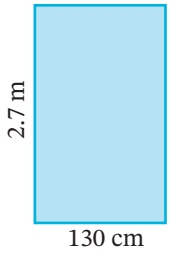
b



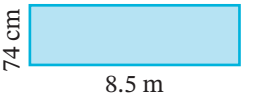
c



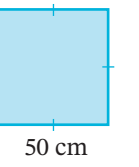
d



e



f



7 The front yard of a house measures 20 m by 8 m while the backyard measures 35 m by 7.5 m. Calculate the cost of covering both yards with grass if it costs $18.60 per square metre. PS

8 A rectangle has a width of 4 m and an area of 18 m^2. What is its length?

9 Sandra's bedroom floor measures 3.6 m by 2.8 m. She buys carpet squares of length 40 cm.

a What is the area of Sandra's bedroom floor?

b What is the area of a carpet square in square metres?

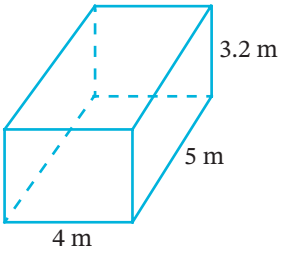
c How many carpet squares are needed to cover Sandra's bedroom floor? PS

10 A rectangle has an area of 35 cm^2. What could its length and width be? R

11 Qiang has a square photo with an area of 81 cm^2. She needed to enlarge the photo to an area of 729 cm^2. To achieve this, each side had to be multiplied by what number? Select **A**, **B**, **C** or **D**. R

A 9 **B** 6 **C** 4 **D** 3

12 A 4 m × 5 m basement has a ceiling height of 3.2 m. PS



a Find the area of:

i the floor

ii the ceiling

iii one of the smaller walls

b How many square tiles measuring 50 cm on each side are needed to cover the floor of the basement?

c How many litres of paint are needed to cover the 4 walls of the basement if one litre of paint covers 16 m^2?

Did you know?

Unusual units of measurement

The Sydharb is an unusual unit used by some Australian government departments to measure a large volume of water. One Sydharb is roughly the amount of water in Sydney harbour, about 562 000 megalitres.

In horse sports, the height of a horse is measured in 'hands'. A hand is 4 inches or 10.2 cm, roughly the width of a human hand.



Shutterstock.com/ashkabe

1 **How many litres of water is one Sydharb?**

2 **What is the average height of a horse in 'hands'?**

3 **What is a 'dog year'?**

8.05

Quiz
Mental skills 8A

Mental skills 8A: Maths without calculators ANSWERS ON P. 588

Doubling and halving numbers

You can double or halve a number by splitting it up first, then doubling or halving.

1 Study each example.

a $92 \times 2 = 180 + 4$
$= 184$
Think: Double 92 = double 90 + double 2

b $37 \times 2 = 60 + 14$
$= 74$
Think: Double 37 = double 30 + double 7

c $\frac{1}{2} \times 86 = 40 + 3$
$= 43$
Think: Half of 86 = half of 80 + half of 6

d $\frac{1}{2} \times 244 = 120 + 2$
$= 122$
Think: Half of 244 = half of 240 + half of 4

e $\frac{1}{2} \times 78 = 30 + 9$
$= 39$
If the tens number is **odd**, then:
Think: Half of 78 = half of 60 + half of 18

f $\frac{1}{2} \times 132 = 60 + 6$
$= 66$
Think: Half of 132 = half of 120 + half of 12

2 Now evaluate each product.

a 54×2	**b** 77×2	**c** 83×2	**d** 105×2
e 26×2	**f** 41×2	**g** 98×2	**h** 162×2
i $\frac{1}{2} \times 182$	**j** $\frac{1}{2} \times 274$	**k** $\frac{1}{2} \times 92$	**l** $\frac{1}{2} \times 138$
m $\frac{1}{2} \times 506$	**n** $\frac{1}{2} \times 76$	**o** $\frac{1}{2} \times 48$	**p** $\frac{1}{2} \times 170$

Foundation | Standard | Complex

Technology

Perimeter and area of a chicken pen

Farmer Jack has 36 m of fencing to make a rectangular chicken pen.

1 Open a new spreadsheet and enter the labels, values and formula for calculating the perimeter of a rectangle as shown below.

	A	B	C	D
1	Chicken Pens			
2				
3	Length (m)	Width (m)	Perimeter (m)	Area (m)
4	10	8	=2*A4+2*B4	
5				

2 In cell D4, enter a formula for the area of the rectangle.

3 Now use your dynamic geometry software to help you solve this problem. Set up your drawing page, check that the grid is showing and the axes are not showing.

4 Construct a rectangle 10 m by 8 m.

5 Now find the area of this rectangle.

6 Now repeat steps **4** and **5** and find at least 3 pairs of dimensions (whole numbers only) for this rectangle and enter them into your spreadsheet.

7 Return to the spreadsheet and enter the dimensions for length and width that you found. Also, **Fill Down** from cells C4 and D4 to complete the perimeter and area calculations.

8 There are more pairs of whole-number dimensions (lengths and widths) for the rectangular chicken pen. Use your dynamic geometry software and fill in your spreadsheet to find any missing pairs of dimensions.

9 Which dimensions make the largest area for the chicken pen?



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Investigation

Area of a triangle

Worksheets
1 cm grid paper
Triangle areas

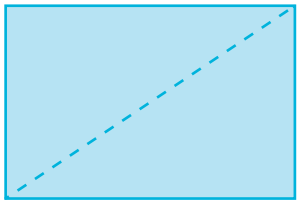
8.05

You will need 1 cm grid paper and scissors.

Right-angled triangles

1 On your grid paper, draw a rectangle 6 cm by 4 cm.

2 Cut the rectangle in half along a diagonal.
What shape have you made?



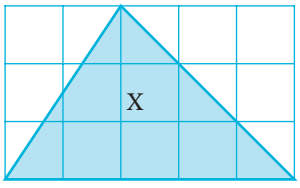
3 Area of rectangle = ____ × ____ = ____cm^2

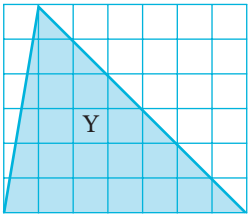
4 So what is the area of each triangle?

5 Draw 2 other rectangles and repeat steps **2** to **4**. What do you notice regarding the area of the triangle and the area of the rectangle?

Non-right-angled triangles

How can we find the area of these triangles?





1 On your grid paper, draw triangle X as shown, and cut out the rectangle.

2 Area of rectangle = _____

3 Cut out triangle X.

4 Place the leftover pieces on triangle X. Do they cover triangle X exactly?

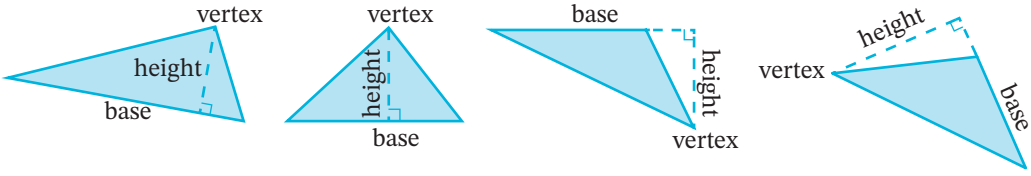
5 Area of triangle = _____

6 Repeat the above steps for triangle Y.

8.06 Area of a triangle

Worksheet
Rectangle and triangle areas

Any side of a triangle can be called its **base**. The **height** is the distance from the base to the opposite vertex. This distance is measured at right angles to the base, so it is also called the **perpendicular height**. Here are some examples:

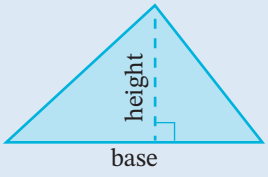


Note that with the last 2 examples involving obtuse-angled triangles, the perpendicular height is **outside** the triangle.

Area of a triangle

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

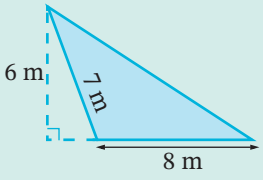
$$A = \frac{1}{2}bh$$



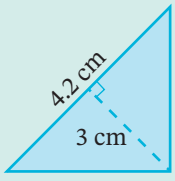
Example 8

Find the area of each triangle.

a



b



Solution

a $\text{Area} = \frac{1}{2}bh$

$= \frac{1}{2} \times 8 \times 6$ Choosing 8 m as the base

$= 24 \text{ m}^2$ The length of 7 m is not required to find this triangle's area

b $\text{Area} = \frac{1}{2}bh$

$= \frac{1}{2} \times 4.2 \times 3$ Choosing 4.2 cm as the base

$= 6.3 \text{ cm}^2$

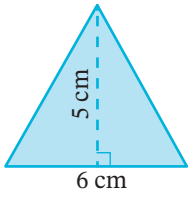
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EXERCISE 8.06 ANSWERS ON P. 589

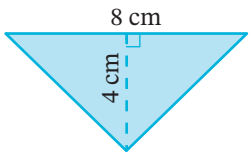
Area of a triangle U F PS R

1 Find the area of each triangle.

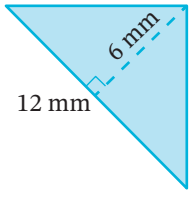
a



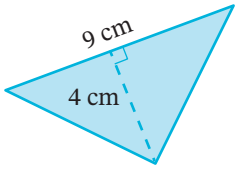
b



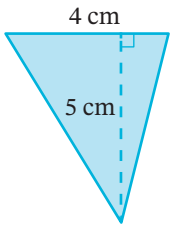
c



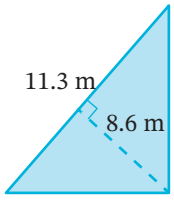
d



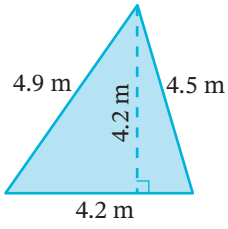
e



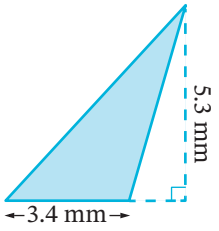
f



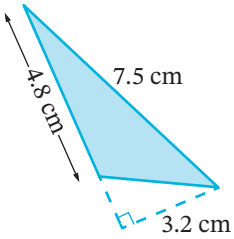
g



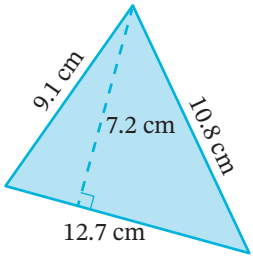
h



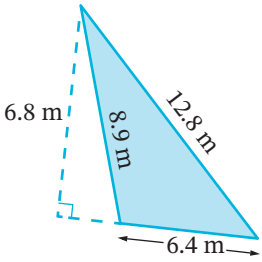
i



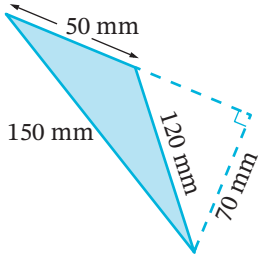
j



k

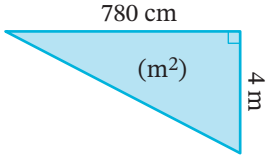


l

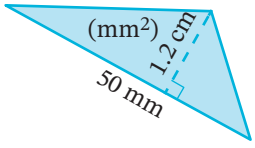


2 Find the area of each triangle, using the units given in brackets.

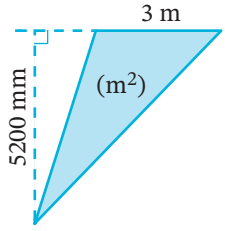
a



b



c

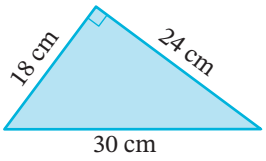


3 A triangle has a base of length 4 m and an area of 40 m^2. What is its perpendicular height? R

4 A triangle with an area of 27 cm^2 has a perpendicular height of 18 cm. How long is its base? R

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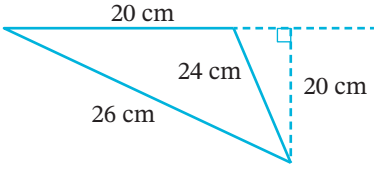
5 What is the area of this triangle?
Select the correct answer **A**, **B**, **C** or **D**.



A 360 cm²
B 432 cm²
C 270 cm²
D 216 cm²

6 The fly-screen of a tent is to be made into the shape of a triangle 0.8 m long and 1.4 m high. Calculate the cost of the screen if one square metre costs $9.10. PS

7 Find the area of this triangle.
Select **A**, **B**, **C** or **D**.

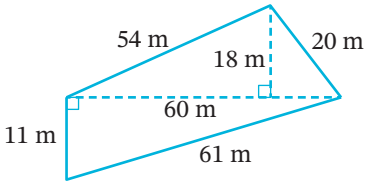


A 130 cm²
B 200 cm²
C 260 cm²
D 240 cm²

8 A triangle has an area of 36 m². What could its base length and perpendicular height be? R

9 The diagram shows the measurements of a new park.

Find the area of the park.



10 A triangle's perpendicular height is the same as its base length. If its area is 32 cm², what is its perpendicular height? PS

Investigation

Area of a parallelogram

A **parallelogram** is a quadrilateral with opposite sides parallel and equal in length.

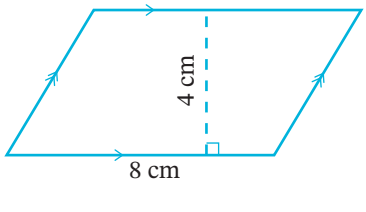
1 On a sheet of paper, draw a parallelogram with a base length of 8 cm and a perpendicular height of 4 cm.

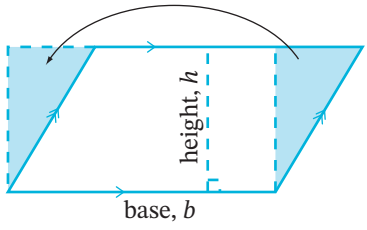
2 Draw an identical parallelogram in your book.

3 Cut out the parallelogram from the sheet of paper.

4 Cut along the dotted line (perpendicular height) and rearrange the pieces to form a rectangle.

5 Measure the length and width of the rectangle to the nearest millimetre, and then find its area.





6 Why is the area of the rectangle the same as the area of the parallelogram?

7 Suggest a general formula for finding the area of any parallelogram. (Check your answer with your teacher.)

8 Paste the rectangle in your book under your diagram of the parallelogram.

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Area of parallelogram

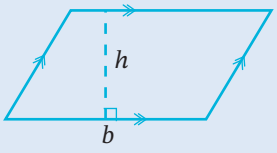
8.07

The area of a **parallelogram** is the same as the area of a rectangle with the same base, b, and perpendicular height, h. It is found by multiplying the base, b, by the height, h.

$$\text{Area} = b \times h = bh$$

Area of a parallelogram

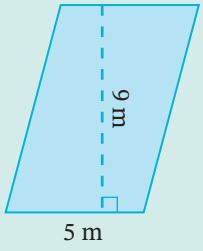
$$A = bh$$



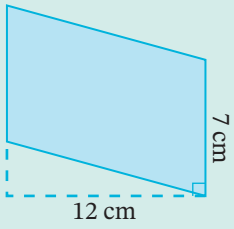
Example 9

Find the area of each parallelogram.

a



b



Solution

a $\text{Area} = bh$

$= 5 \times 9$

$= 45 \text{ m}^2$

b $\text{Area} = bh$

$= 7 \times 12$

$= 84 \text{ cm}^2$

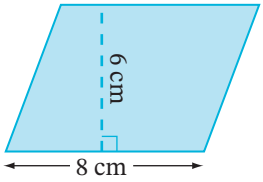
EXERCISE 8.07 ANSWERS ON P. 589

Area of a parallelogram UFR

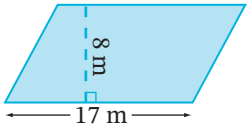
EXAMPLE 9

1 Find the area of each parallelogram.

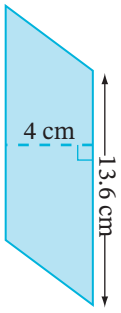
a



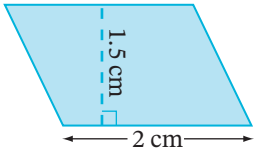
b



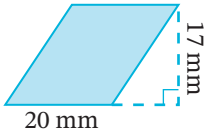
c



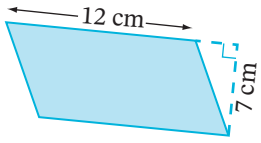
d



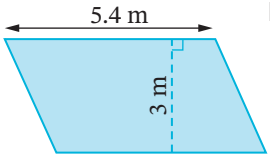
e



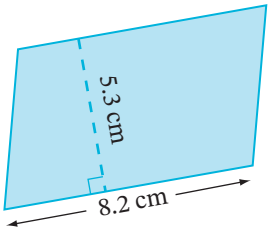
f



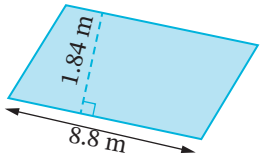
g



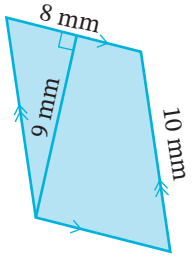
h



i



2 What is the area of this parallelogram? Select the correct answer **A**, **B**, **C** or **D**.



A 72 mm^2

B 36 mm^2

C 45 mm^2

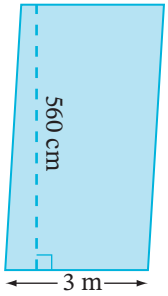
D 80 mm^2

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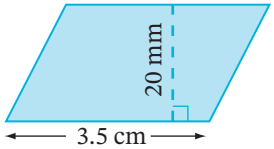
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3 Find the area of each parallelogram in square centimetres. Make sure you change both lengths to centimetres first.

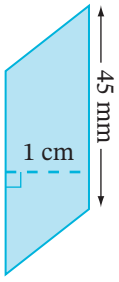
a



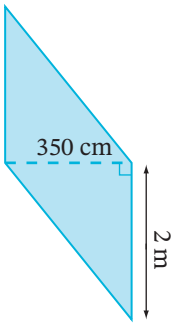
b



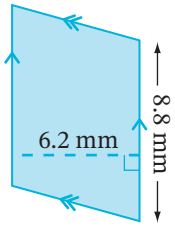
c



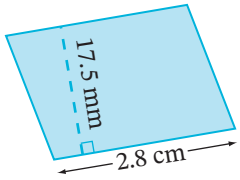
d



e



f



4 The courtyard of a house has the shape of a parallelogram with a base length of 4.3 m and an area of 24.94 m^2. What is its perpendicular height? Select **A**, **B**, **C** or **D**. R

A 5.8 m **B** 10.6 m **C** 53.621 m **D** 107.24 m

5 A parallelogram has an area of 245 cm^2. What could its base length and perpendicular height be? R

6 The area of a parallelogram is the same as the area of a square with side length 30 cm. Find the height of the parallelogram if the base is 36 cm. R



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8.08 Areas of composite shapes

Worksheets
A page of composite shapes

Composite shapes

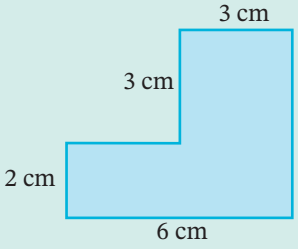
Puzzle
Areas 1

Video
Areas of composite shapes

A **composite shape** is made up of 2 or more shapes.

Example 10

Find the area of this shape.



Solution

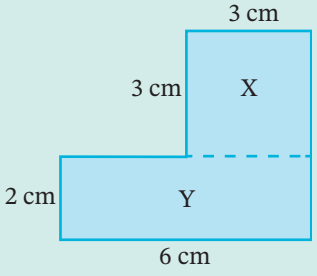
Method 1

Separate the shape into a square (X) and a rectangle (Y).

Area of shape = area of square X + area of rectangle Y

$= 3 \times 3 + 6 \times 2$

$= 21 \text{ cm}^2$



Method 2

Separate the shape into 2 rectangles P and Q.

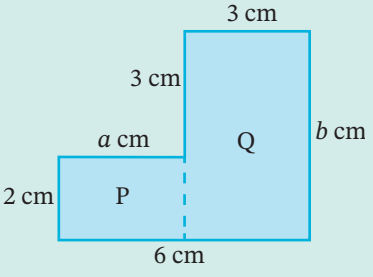
$a = 6 - 3 = 3$

$b = 3 + 2 = 5$

Area of shape = area of P + area of Q

$= 3 \times 2 + 5 \times 3$

$= 21 \text{ cm}^2$



Method 3

The area can also be found by subtracting areas.

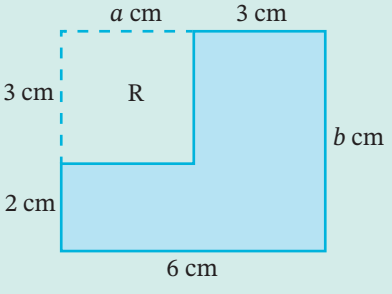
$a = 6 - 3 = 3$

$b = 3 + 2 = 5$

Area of shape = area of big rectangle – area of R

$= 6 \times 5 - 3 \times 3$

$= 21 \text{ cm}^2$

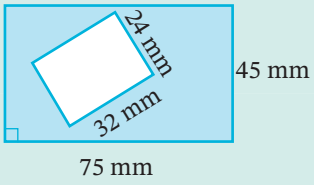


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Example 11

Find the shaded area.



Solution

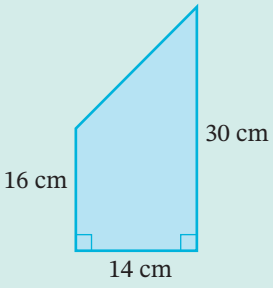
Shaded area = area of big rectangle – area of small rectangle

$= 75 \times 45 - 32 \times 24$

$= 2607 \text{ mm}^2$

Example 12

Find the area of this shape.



Solution

Divide the shape into a triangle and a rectangle.

Height of triangle = 30 – 16

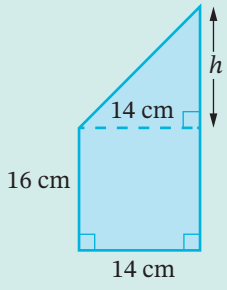
= 14 cm

Area of shape = area of rectangle + area of triangle

$= 16 \times 14 + \frac{1}{2} \times 14 \times 14$

$= 224 + 98$

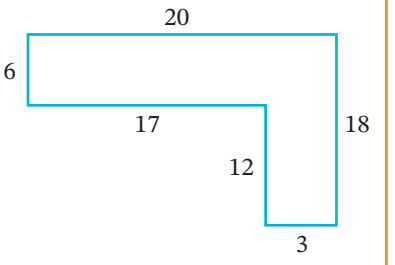
$= 322 \text{ cm}^2$



EXERCISE 8.08 ANSWERS ON P. 589

Areas of composite shapes U F PS R

1 Which expression can be used to calculate the area of this shape? Select the correct answer **A**, **B**, **C** or **D**.



A $(20 \times 18) - (17 \times 12)$

B $(18 \times 3) + (20 \times 6)$

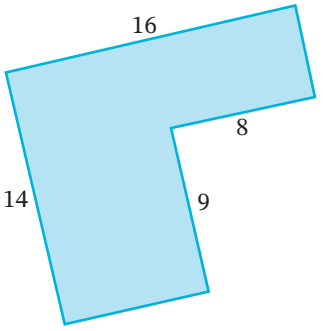
C $(20 \times 18) - (17 \times 6)$

D $(18 \times 3) + (17 \times 3)$

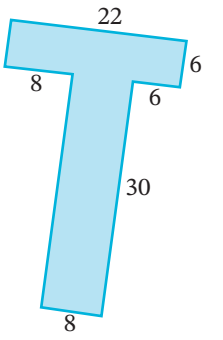
EXAMPLE 10

2 Find the area of each shape. All measurements are in centimetres. PS R

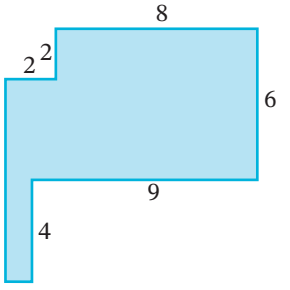
a



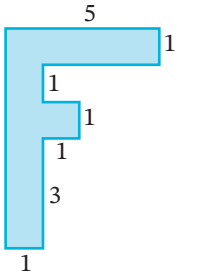
b



c



d



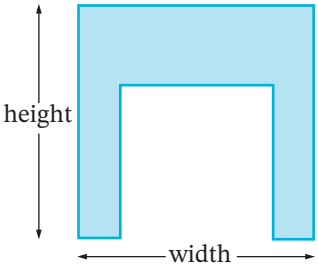
EXAMPLE 11

3 Find the area of each shape. PS R

a Outside dimensions: 230 cm × 55 cm.
Hole dimensions: 40 cm × 35 cm



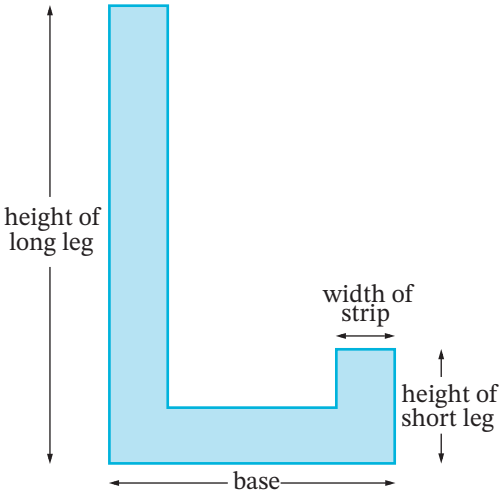
b Total height: 30 cm, total width: 30 cm
Dimensions of cut-out: 20 cm × 20 cm



Foundation Standard Complex

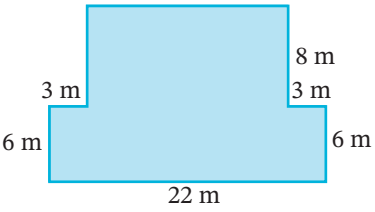
9780170477963

c This shape is formed from strips of metal 10 cm wide. The height of the long leg is 80 cm. The base is 50 cm wide. The height of the short leg is 20 cm.



EXAMPLE 12

4 This diagram shows a swimming pool at a holiday resort. What is the area of the floor of the pool? Select **A**, **B**, **C** or **D**. PS R

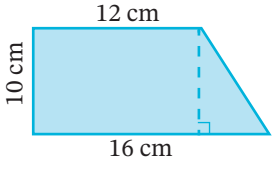


A 156 m^2 **B** 180 m^2

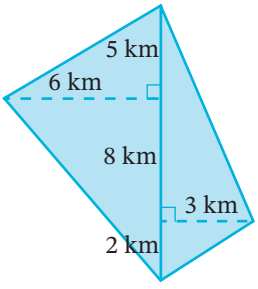
C 72 m^2 **D** 260 m^2

5 Find the area of each shape. PS R

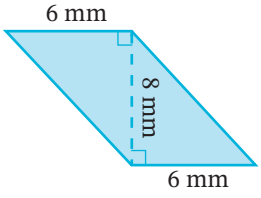
a



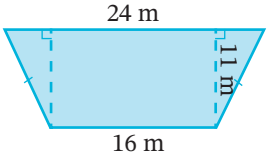
b



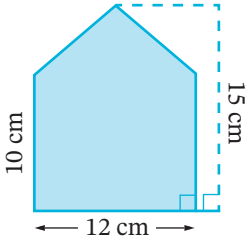
c



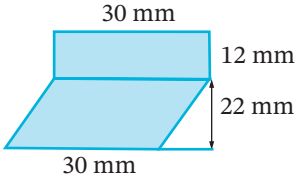
d



e



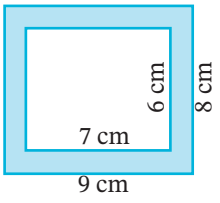
f



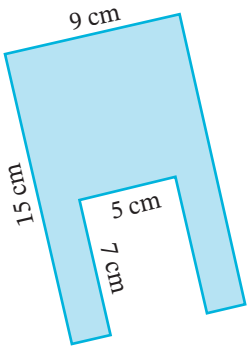
Foundation Standard Complex

6 Find each shaded area. PS R

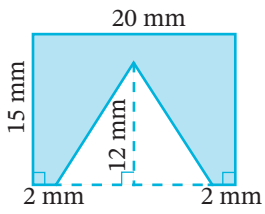
a



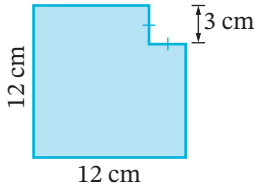
b



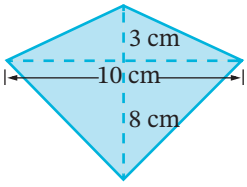
c



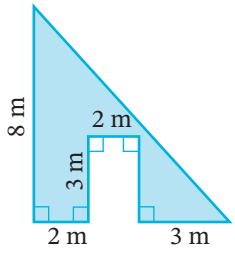
d



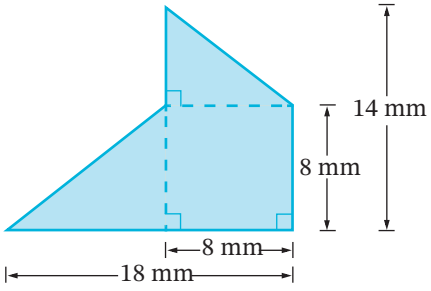
e



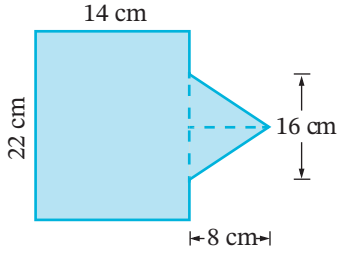
f



g

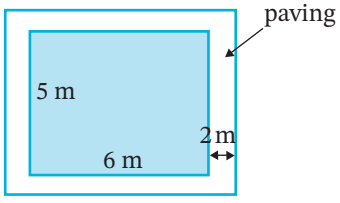


h

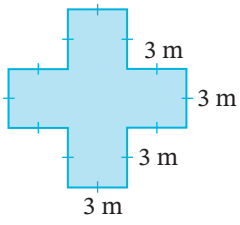


7 A rectangular lawn contains 2 square flower gardens, each measuring 3 m by 3 m. If the lawn measures 9.5 m by 6.4 m, find the area of the grass. PS R

8 A garden measuring 6 m by 5 m has a 2 m wide strip of paving around its border. What is the area of the paving? PS R



9 The floor of a children's pool is in the shape of a cross as shown. Each side is 300 cm long. How many tiles are required to tile this floor if one tile covers 1600 cm^2? PS R



Foundation Standard Complex

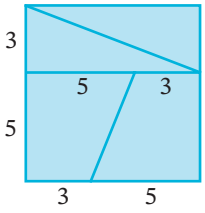
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Investigation

Area puzzles

This investigation can also be printed out as the puzzle sheet 'Area puzzles' to save you drawing the diagrams.

1 Draw this square on 1 cm dot paper or use the square from 'Area puzzles'. All measurements are in centimetres. What is the area of this square in cm^2?

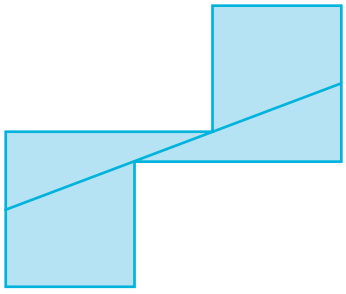


2 **a** Cut out the square, then cut along the marked lines.

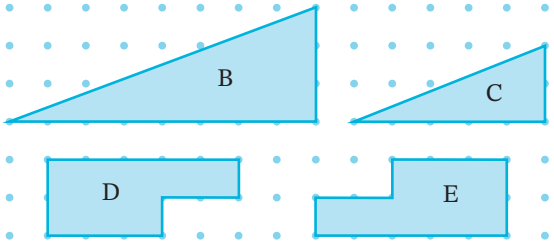
b Arrange the 4 pieces to make a rectangle. Find its length and width, and use these to find its area.

3 Is the calculated area of the rectangle the same as the area of the original square? Should it be? Explain this mystery.

4 Use the same 4 pieces to make this shape. Calculate its area. Is this the same as the area of the original square? Should it be? Why?

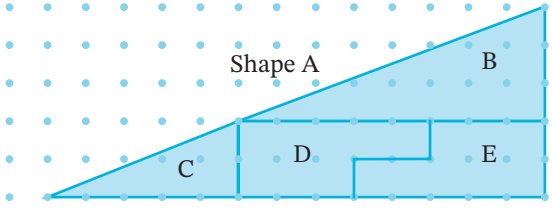


5 **a** Use square dot paper to draw each of the shapes B, C, D and E as shown, or use the shapes on the sheet 'Area puzzles'. Find the area of each shape in cm^2.



b Calculate the sum of the 4 areas.

c Cut out the 4 shapes and combine them to form Shape A as shown below.



6 **a** Shape A looks like a right-angled triangle. Find its height and the length of its base, and use them to find its area.

b Does the area of the triangle equal the sum of the areas of B, C, D and E from question **1**? Explain the mystery about the area of the triangle, the area of Shape A and the sum of the areas of B, C, D and E.

Puzzle
Area puzzles

Worksheet
Square dot paper

8.08

Technology

Local areas

1 **a** Use an online map such as Google Maps to find a park, garden or reserve near your home.

b Find the scale for measuring distance used on your map.

c Estimate the area of the park, garden or reserve.

2 **a** Use the map's scale to make a grid of squares on your map, either on screen or on a printout of your map.

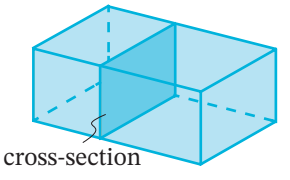
b Approximate the area (in m^2 and ha) of your park, garden or reserve. Use the method of counting squares. (It may be more accurate to subdivide each square on the grid into smaller shapes.)

8.09 Drawing prisms

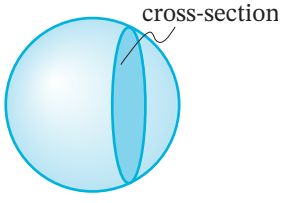
Videos
Escher and the endless staircase
Perspective: Dazzle camouflage

Cross-sections

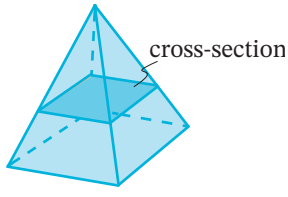
A **cross-section** of a solid is a 'slice' of the solid, cut *across* it, parallel to its end faces, rather than along it. These diagrams show cross-sections of a rectangular prism, a sphere and a square pyramid.



Rectangular prism



Sphere



Square pyramid

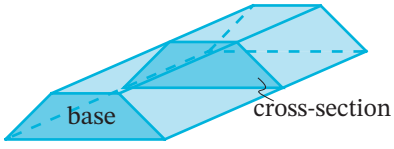
Note that the cross-sections of a rectangular prism are congruent (identical) rectangles, the cross-sections of a sphere are circles (but of different sizes), and the cross-sections of a square pyramid are squares (again, of different sizes).

Prisms

If a solid has the same (uniform) cross-section along its length, and each cross-section has straight sides (not rounded), then the solid is called a **prism**. So, from the above diagrams:

- a rectangular prism is a prism (its cross-sections are uniform rectangles)
- a sphere is not a prism (its cross-sections are not uniform)
- a square pyramid is not a prism (its cross-sections are not uniform)

The end faces of a prism are called its **base**. Prisms take their names from their base. For example, this is a **trapezoidal prism** because its cross-sections are all trapeziums, identical and parallel to its base.



Trapezoidal prism

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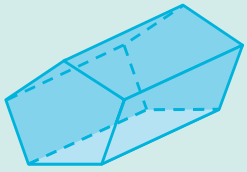
Example 13

a Draw a cross-section of the prism shown.

b What shape is the prism's base?

c What shape are the side faces?

d What is the name of this prism?



Solution

a



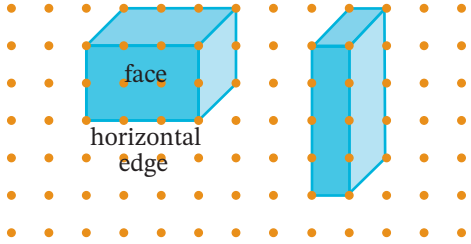
b The base is a pentagon.

c The side faces are all rectangles.

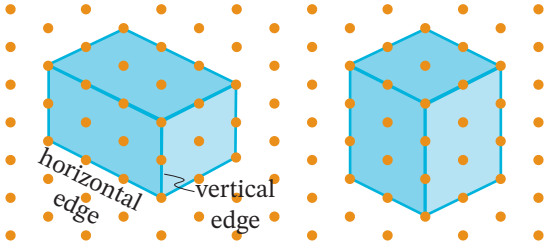
d This is a pentagonal prism.

Drawing different views of solids

Prisms can be difficult to draw. Sometimes it is easier to use square or isometric dot paper to draw them. These rectangular prisms are drawn on **square dot paper**, a square grid of dots.



The rectangular prisms below are drawn on **isometric dot paper**, a '3D' triangular grid of dots arranged as equilateral triangles. A 2D (flat) diagram of a 3D figure (solid) looks more natural when drawn on isometric paper. On isometric paper, all horizontal edges are shown at an angle of 30°.



- When drawing on **square** paper, one of the prism's **faces** is shown at the forefront.
- When drawing on **isometric** paper, one of the prism's **vertical edges** is shown at the forefront.

Example 14

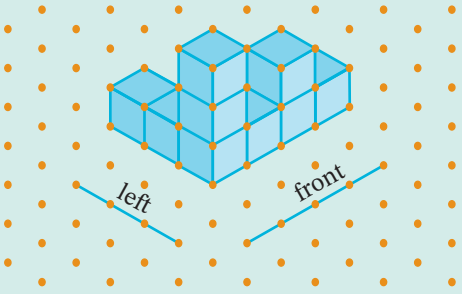
For the solid shown, assume that there are no hidden cubes. Draw:

a the front view

b the left view

c the top view

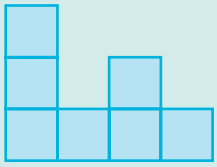
It may help your drawing if you built these solids using centicubes first.



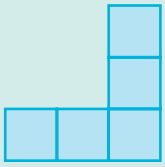
Solution

For each view, there will be some cubes hidden behind the cubes closest to you.

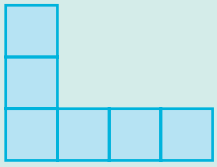
a Viewed from the front



b Viewed from the left



c Viewed from the top



Example 15

For the solid shown, draw:

a the front view

b the top view



Solution

a Viewed from the front



b Viewed from the top



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EXERCISE 8.09 ANSWERS ON P. 589

Drawing prisms U F R C

For this exercise you will need square dot paper, isometric dot paper and centicubes.
Both types of dot paper can be printed out from Nelson MindTap. You can also draw shapes using dynamic geometry software, using a square grid or an isometric grid.

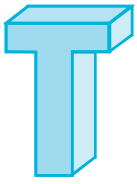
Worksheets
Square dot paper
Isometric dot paper

8.09

1 State whether each solid is a prism or not. R C

a 

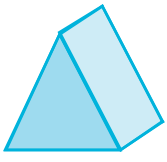
b 

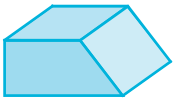
c 

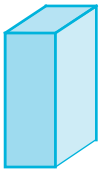
d 

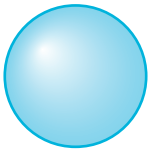
e 

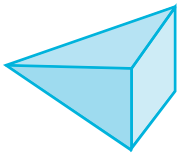
f 

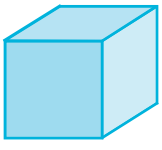
g 

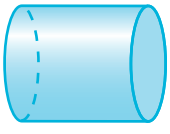
h 

i 

j 

k 

l 

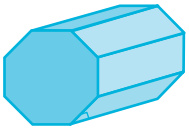
m 

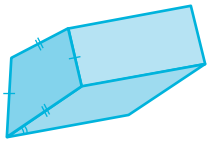
n 

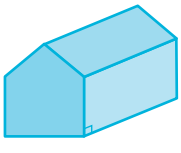
EXAMPLE 13

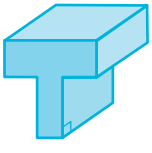
2 For each prism: R C

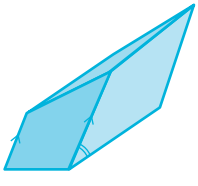
i draw its cross-section **ii** write the name of the prism

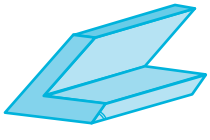
a 

b 

c 

d 

e 

f 

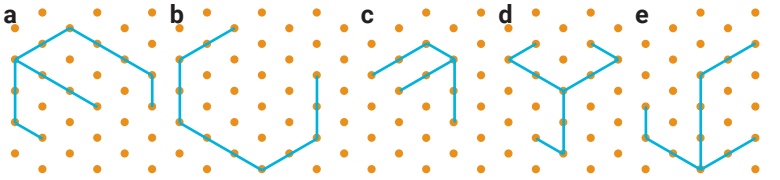
3 Draw each prism and shade its base.

a square prism **b** isosceles triangular prism

c trapezoidal prism **d** hexagonal prism

Foundation Standard Complex

4 Each diagram is the start of a drawing of a rectangular prism. Copy each drawing onto isometric dot paper and complete it.

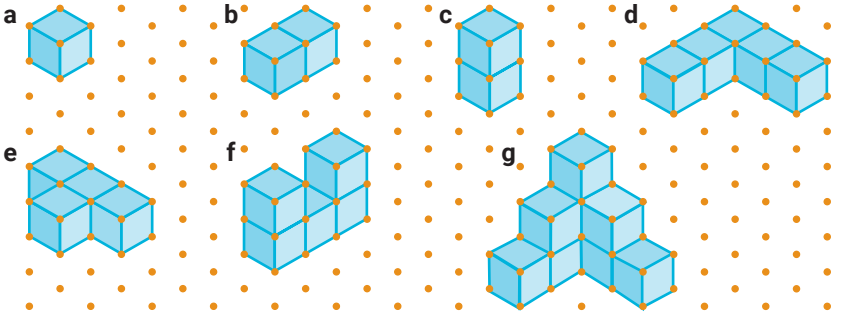


5 Draw a triangular prism:

a on square dot paper

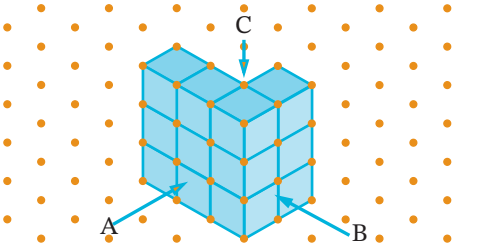
b on isometric dot paper.

6 Use centicubes to make each solid, then draw them on isometric dot paper.

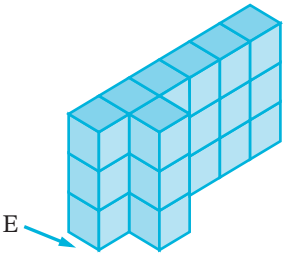


EXAMPLE 14

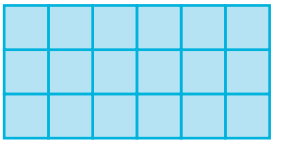
7 Draw what you would see of this prism if you were looking from A, then from B, and then from C. R C



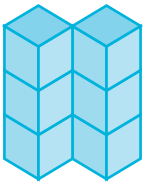
8 What would Emma see if she were looking at this prism from point E? Select the correct answer **A**, **B**, **C** or **D**.



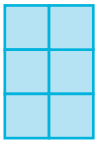
A



B



C



D

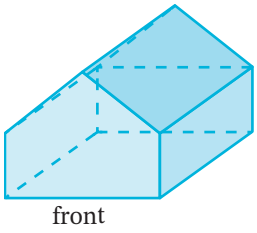


Foundation Standard Complex

9780170477963

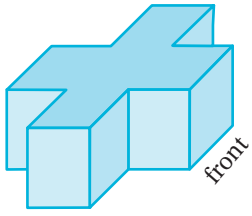
9 For each prism, draw each view requested. R C

a



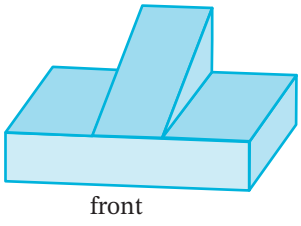
i front view
ii right view

b



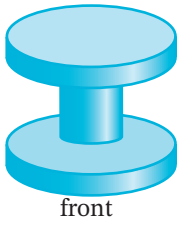
i left view
ii top view
iii front view

c



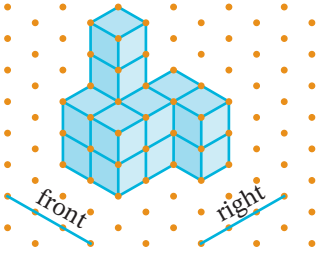
i front view
ii left view
iii right view
iv top view

d



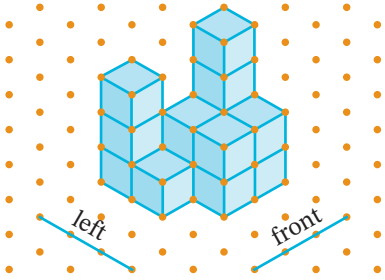
i top view
ii front view

e



i front view
ii right view
iii top view

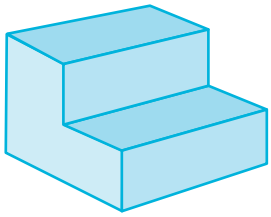
f



i back view
ii left view
iii top view

10 Damien flew over the top of this building.

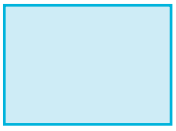
Which view would he see when directly overhead?
Select **A**, **B**, **C** or **D**.



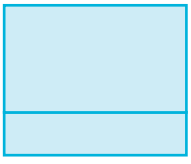
A



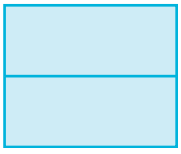
B



C



D



Quiz
Mental skills 8B

Mental skills 8B: Maths without calculators ANSWERS ON P. 589

Multiplying and dividing by 4 or 8

Multiplying and dividing by 4 or 8 involves repeated doubling or halving.

1 Study each example.

a 65×4 — Think: Double twice.
Double $65 = 130$
Double $130 = 260$

b 27×4 — Think: Double twice.
Double $27 = 54$
Double $54 = 108$

c 14×8 — Think: Double 3 times.
Double $14 = 28$
Double $28 = 56$
Double $56 = 112$

d $236 \div 4$ — Think: Halve twice.
$\frac{1}{2} \times 236 = 118$
$\frac{1}{2} \times 118 = 59$

e $564 \div 4$ — Think: Halve twice.
$\frac{1}{2} \times 564 = 282$
$\frac{1}{2} \times 282 = 141$

f $392 \div 8$ — Think: Halve 3 times
$\frac{1}{2} \times 392 = 196$
$\frac{1}{2} \times 196 = 98$
$\frac{1}{2} \times 98 = 49$

2 Now evaluate each expression.

a	14×4	**b**	27×4	**c**	16×4	**d**	105×4
e	43×8	**f**	16×8	**g**	28×8	**h**	33×8
i	$184 \div 4$	**j**	$272 \div 4$	**k**	$560 \div 4$	**l**	$432 \div 4$
m	$624 \div 8$	**n**	$312 \div 8$	**o**	$256 \div 8$	**p**	$152 \div 8$

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Metric units for volume

8.10

Volume

The **volume** of a solid is the amount of space taken up by the solid.

Skillsheet
What is volume?

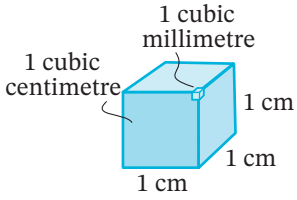
8.10

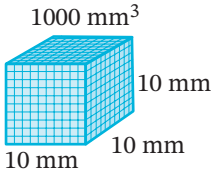
Video
Metric units of volume

The table shows metric units of volume.

Volume unit	Abbreviation	The size of a cube with each side measuring ...
cubic millimetre	mm^3	1 mm
cubic centimetre	cm^3	1 cm
cubic metre	m^3	1 m

A **cubic millimetre** is about the size of a grain of raw sugar or rock salt. A **cubic centimetre** is about the size of a person's tooth or a medical pill.

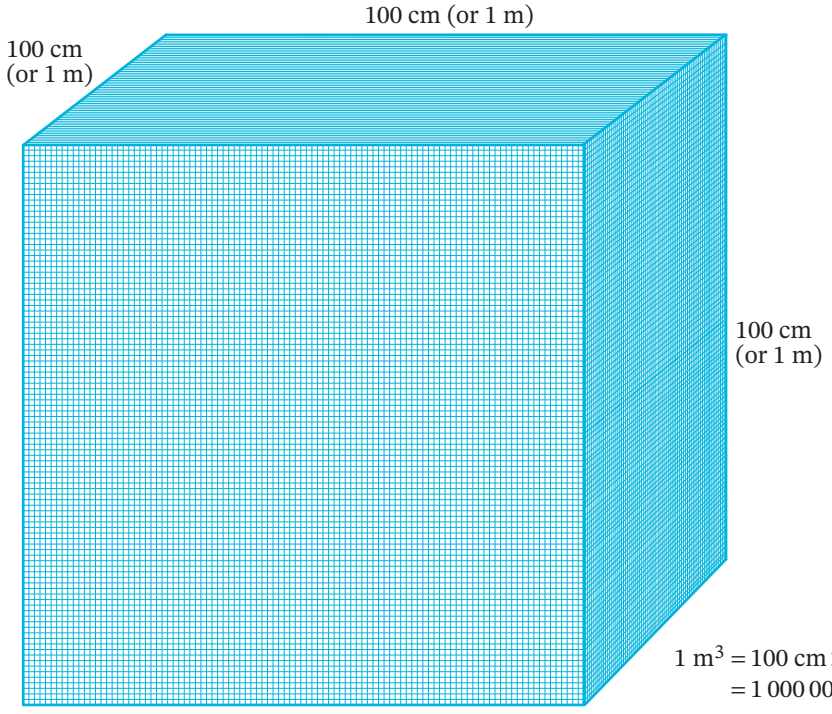




$1\ cm^3 = 10\ mm \times 10\ mm \times 10\ mm$
$= 1000\ mm^3$

There are 1000 cubic millimetres in one cubic centimetre.

A **cubic metre** is about the size of 2 washing machines. There are 1 000 000 cubic centimetres in one cubic metre.



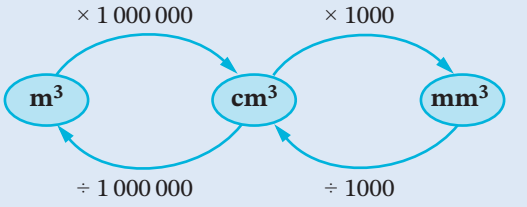
$1\ m^3 = 100\ cm \times 100\ cm \times 100\ cm$
$= 1\,000\,000\ cm^3$

Metric units for volume

$1\ cm^3 = 1000\ mm^3$

$1\ m^3 = 1\ 000\ 000\ cm^3$

This diagram shows how to convert between different units of volume.



Example 16

Convert:

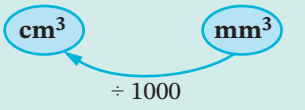
a $12\ 000\ mm^3$ to cm

b $48\ m^3$ to cm^3

Solution

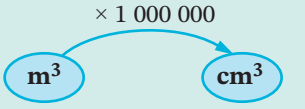
a $12\ 000\ mm^3 = (12\ 000 \div 1000)\ cm^3$

$= 12\ cm^3$



b $48\ m^3 = (48 \times 1\ 000\ 000)\ cm^3$

$= 48\ 000\ 000\ cm^3$



EXERCISE 8.10 ANSWERS ON P. 589

Metric units for volume U F C

1 What unit of volume would you use to measure the volume of: C

a a textbook?

b a backpack?

c a wardrobe?

d a large suitcase?

e a matchbox?

f a room?

2 What is the approximate volume of a brick? Select the correct answer **A**, **B**, **C** or **D**.

A $20\ cm^3$

B $1000\ cm^3$

C $1600\ cm^3$

D $3000\ cm^3$

3 Copy and complete each conversion.

a $3\ cm^3 =$ _____ mm^3
b $5\ m^3 =$ _____ cm^3
c $2.6\ cm^3 =$ _____ mm^3
d $4000\ mm^3 =$ _____ cm^3
e $7.2\ m^3 =$ _____ cm^3
f $66\ 000\ mm^3 =$ _____ cm^3
g $1\ m^3 =$ ____ mm^3
h $2300\ cm^3 =$ ____ m^3
i $126\ 000\ 000\ cm^3 =$ ___ m^3
j $3450\ mm^3 =$ _____ cm^3
k $25\ m^3 =$ _____ mm^3
l $78\ 000\ mm^3 =$ _____ m^3
m $63\ 000\ cm^3 =$ _____ m^3
n $1.4\ mm^3 =$ _____ cm^3
o $5000\ cm^3 =$ _____ m^3

4 The volume of a chest of drawers is $306\ 000\ cm^3$. Convert this to cubic metres. C

5 Match the correct volume (**A** to **G**) with each of the items (**a** to **g**) listed. C


Shutterstock.com/Brent Hofacker



Shutterstock.com/Gts

a a bottle of liquid paper
b a box of tissues
c a glass of water
d a bottle of lemonade
e a classroom
f a school hall
g a cereal package
h a tall carton of milk

A $200\ m^3$
B $3890\ m^3$
C $1250\ cm^3$
D $5000\ cm^3$
E $20\ 000\ mm^3$
F $250\ cm^3$
G $2200\ cm^3$
H $1000\ cm^3$

6 The volume of a lunchbox is $2520\ cm^3$. Convert this to cubic millimetres. C

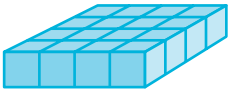
7 A swimming pool has a volume of $38\ m^3$. Convert this to cubic centimetres. C

Investigation

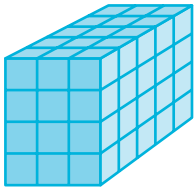
Volume of a rectangular prism

1 These rectangular prisms are made up of 1 cm cubes. Copy and complete the table.

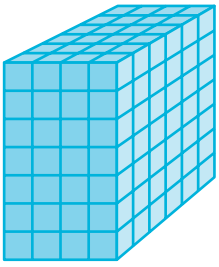
a



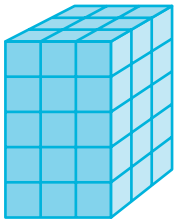
b



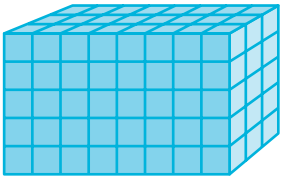
c



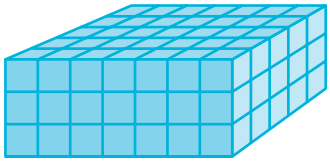
d



e

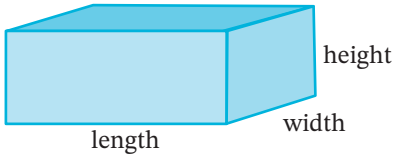


f



Shape	Number of cubes in one layer	Number of layers	Volume (cm^3)
a			
b			
c			
d			
e			
f			

2 Copy and complete this table for the rectangular prisms in question **1**.



Shape	Length (cm)	Width (cm)	Height (cm)	Volume (cm^3)
a	4	4	1	16
b				
c				
d				
e				
f				

3 What is the relationship between the length, width and height of a rectangular prism and its volume?

4 Write the relationship as a rule:
Volume of a rectangular prism = ____ × ____ × ___

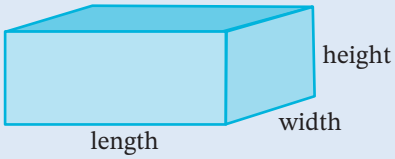
9780170477963

Volume of a rectangular prism 8.11

Volume of a rectangular prism

Volume = length × width × height

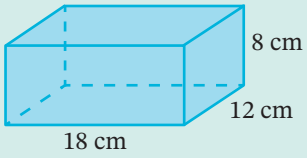
$V = lwh$



Worksheet
Volume
8.11

Example 17

Find the volume of the rectangular prism.



Solution

$V = lwh$

$= 18 \times 12 \times 8$

$= 1728$

The volume is 1728 cm^3.

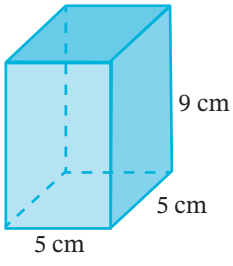
EXERCISE 8.11 ANSWERS ON P. 590

Volume of a rectangular prism U F R

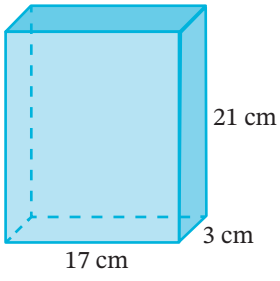
1 Find the volume of each rectangular prism.

EXAMPLE 17

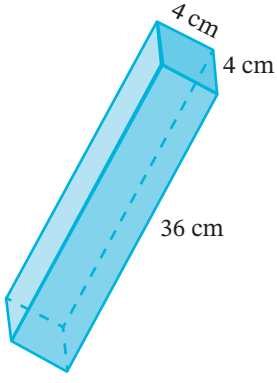
a



b

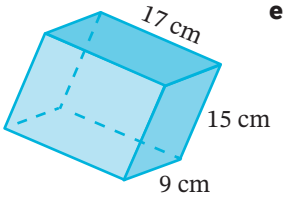


c

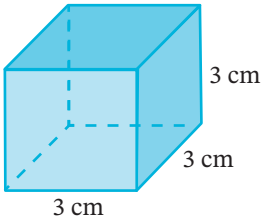


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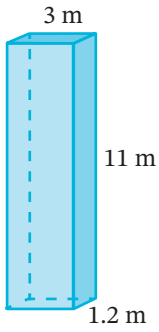
d



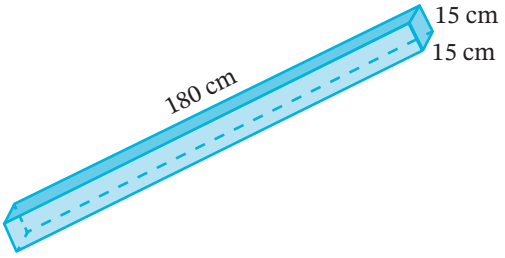
e



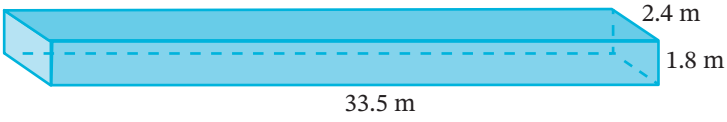
f



g



h



2 Calculate the volume of each rectangular prism described.

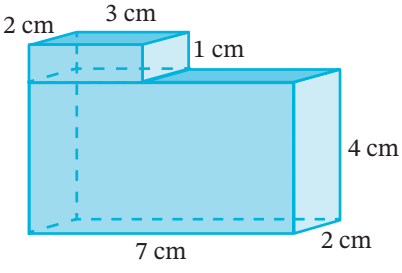
	Length	Width	Height
a	50 cm	50 cm	50 cm
b	5 cm	10 cm	18 cm
c	4 m	2.5 m	1.4 m
d	24 mm	16 mm	11 mm

3 A tissue box has dimensions of 21 cm by 9 cm by 12 cm. Calculate its volume:

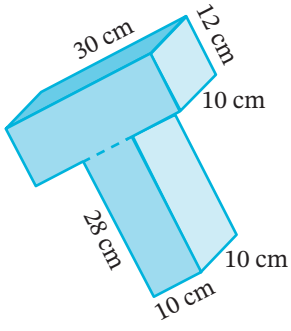
a in cubic centimetres **b** in cubic millimetres

4 Find the volume of each composite prism.

a



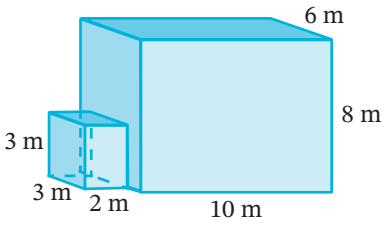
b



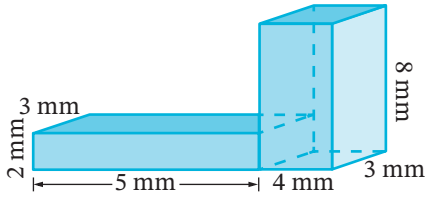
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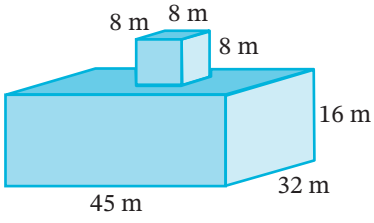
c



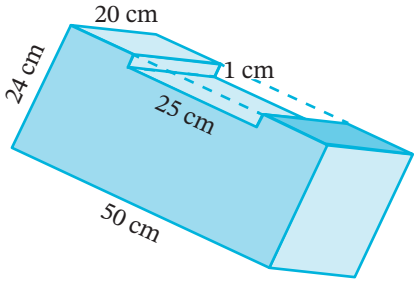
d



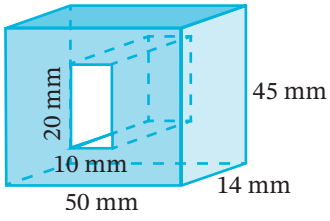
e



f



g

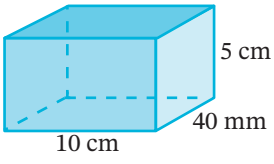


5 A refrigerator has dimensions of 68 cm by 166 cm by 65 cm. Calculate its volume:

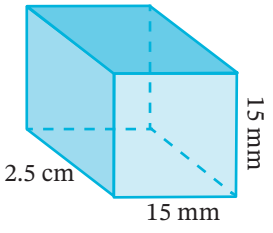
a in cubic centimetres **b** in cubic metres

6 Find the volume of each rectangular prism in cubic centimetres. Make sure you change all measurements to cm first.

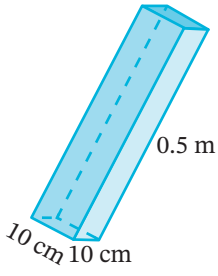
a



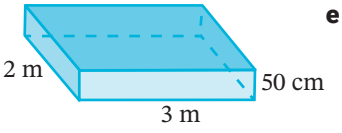
b



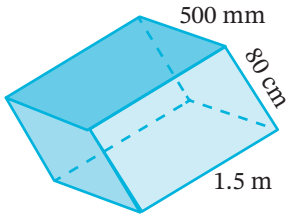
c



d



e



7 A rectangular prism has length 5 mm, width 3 mm and volume 100 mm^3. What is its height as a mixed numeral? R

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8 A rectangular prism has width 1.5 m, height 3 m and volume 27 m^3. What is its length? R

9 A rectangular prism has a volume of 880 cm^3. What could its length, width and height be?

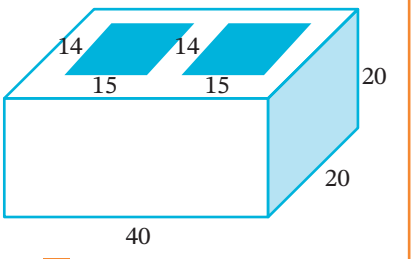
10 A rectangular garden is 12 metres long and 4 metres wide. It is filled with soil to a depth of 15 centimetres. Calculate the volume of the soil. Select the correct answer **A**, **B**, **C** or **D**.

A 0.72 m^3 **B** 7.2 m^3 **C** 72 m^3 **D** 720 m^3

11 Calculate the volume of the hollow cement block. All dimensions are in centimetres. Select **A**, **B**, **C** or **D**.

A 7600 cm^3 **B** 11 800 cm^3

C 15 580 cm^3 **D** 16 000 cm^3

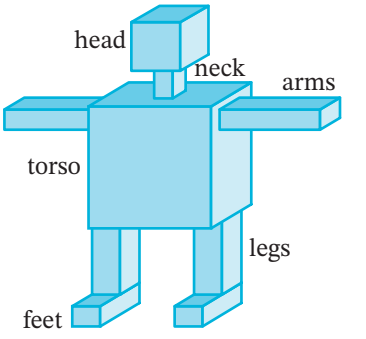


12 A cube has a volume of 125 m^3. What is its side length? R

Investigation

What is your volume?

Imagine that you are made up of rectangular prisms.



1. With the help of a partner, make measurements of your body. Use them to find dimensions (to the nearest centimetre) for each of the prism body parts.
2. Sketch each body part prism and label its dimensions.
3. Use the prisms to find your volume, in cm^3.
4. Write a report of what you did, showing all diagrams and calculations. Explain how you found the dimensions (length, width and height) for the prisms. Do you believe that you found a good approximation of your volume? Why?

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Volume of a prism

8.12

Because a prism is made up of identical cross-section 'slices' along its length, its volume can be calculated by multiplying the **area of the cross-section** by its **perpendicular height** (the length or depth of the prism).

Skillsheet Solid shapes

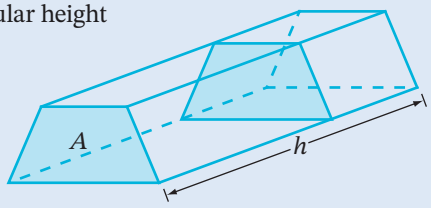
Worksheet Measuring shapes review

Video Volume of a prism

Volume of a prism

Volume of a prism = area of cross-section × perpendicular height

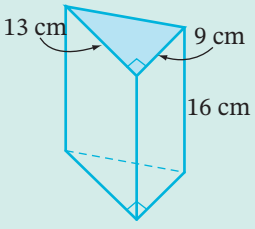
$= Ah$



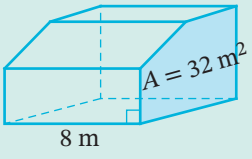
Example 18

Find the volume of each prism.

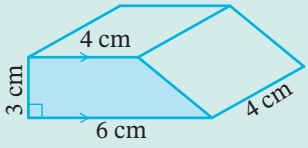
a



b



c



Solution

a $A = \frac{1}{2} \times 9 \times 13$ — Area of shaded triangular cross-section

$= 58.5 \text{ cm}^2$

$V = 58.5 \times 16$ — $V = Ah$, where height $h = 16$ cm

$= 936 \text{ cm}^3$

b $A = 32 \text{ m}^2$ — Area of shaded cross-section

$V = 32 \times 8$ — $V = Ah$, where height $h = 8$

$= 256 \text{ m}^3$

c $A = 4 \times 3 + \frac{1}{2} \times 2 \times 3$ — Cross-section is a rectangle + triangle with base 2 cm

$= 15 \text{ cm}^2$

$V = 15 \times 4$ — $V = Ah$, where height $h = 4$

$= 60 \text{ cm}^3$

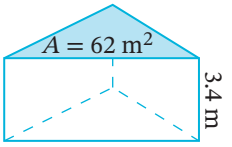
EXERCISE 8.12 ANSWERS ON P. 590

Volume of a prism UFR

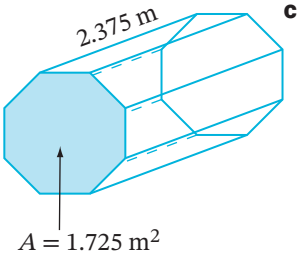
EXAMPLE 18

1 Find the volume of each prism.

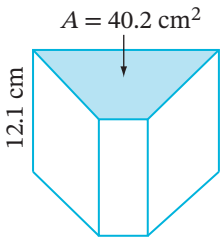
a



b

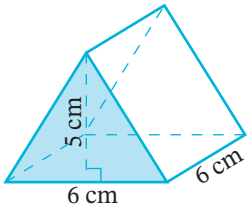


c

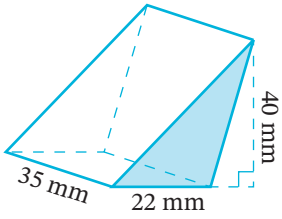


2 Find the volume of each triangular prism.

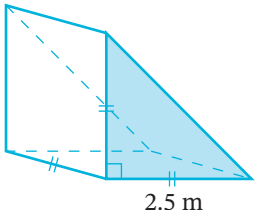
a



b

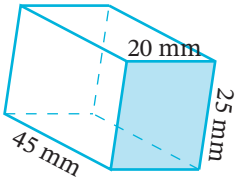


c

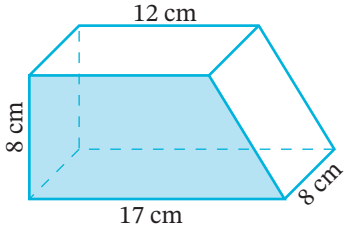


3 Find the volume of each prism.

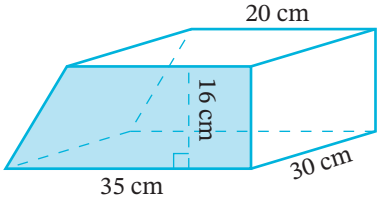
a



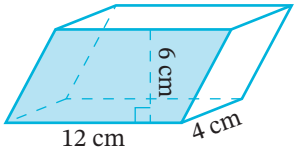
b



c



d

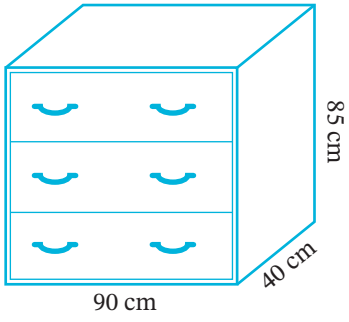


4 Find the volume of this chest of drawers.
Give the answer in m^3:

a by calculating the volume in cm^3, then converting to m^3

b by converting each length to metres first, then calculating the volume.

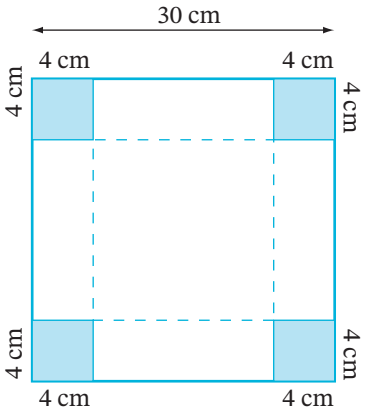
Which method is easier?



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5 A square sheet of metal has a side length of 30 cm. 4 identical squares of length 4 cm are cut away from the corners, as shown. What is the volume of the prism formed when the remaining shape is folded along the dotted lines? Select **A**, **B**, **C** or **D**. R



A 1936 cm³ **B** 2704 cm³

C 64 cm³ **D** 3600 cm³

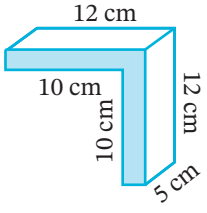
6 A triangular prism has base length 6 m, height 10 m and volume 150 m³. What is its length? R

7 A cube has a volume of 343 m³. What is its side length? R

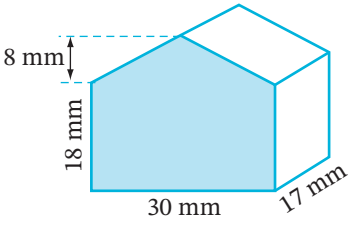
8 A triangular prism has a volume of 36 cm³. What could its length, perpendicular height and length be? R

9 Calculate the volume of each prism.

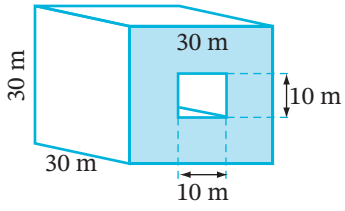
a



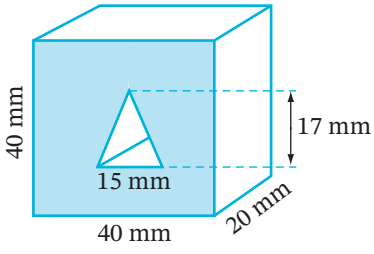
b



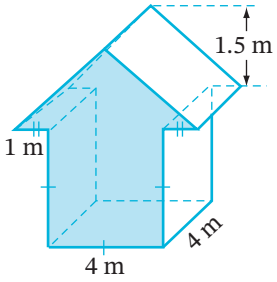
c



d



e



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8.13 Volume and capacity

‘What is the capacity of the water tank?’

Capacity is the amount of fluid (liquid or gas) in a container.

The metric units of capacity are the litre (L), the millilitre (mL), the kilolitre (kL) and the megalitre (ML).

- A large drop of water is about 1 mL
- A teaspoon holds about 5 mL
- A tall standard bottle of milk holds 1 L
- A small rainwater tank holds about 1 kL
- It takes 40 minutes to pump 1 kL of water out of a garden hose
- Half an Olympic-sized swimming pool holds about 1 ML

Worksheets
Volume and capacity cards
Volume and capacity
Capacity

Puzzle
Capacity puzzle

Video
Volume and capacity

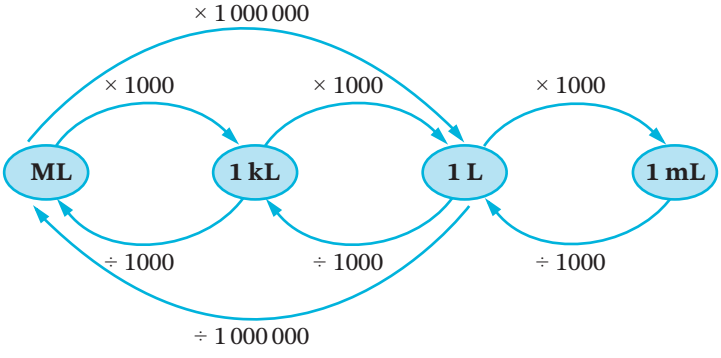
Metric units for capacity

1 L = 1000 mL

1 kL = 1000 L

1 ML = 1000 kL = 1 000 000 L

This diagram shows how to convert between different units of capacity.



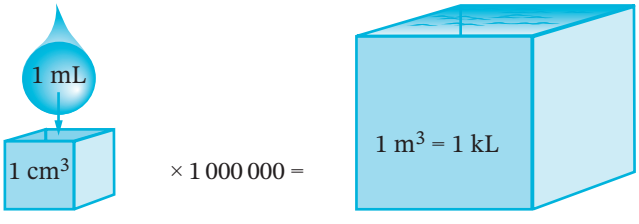
It is also useful to know the relationship between **volume and capacity**.

Volume and capacity

1 cm^3 contains 1 mL

1 m^3 contains 1 kL or 1000 L

This means that a cubic centimetre can hold 1 mL of liquid, while a cubic metre can hold 1000 L of liquid.



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EXERCISE 8.13 ANSWERS ON P. 590

Volume and capacity U F PS R C

1 State what unit of capacity you would use when measuring: C

a a glass of milk
b a dam
c a car's petrol tank
d a bottle of medicine
e an office water cooler
f a backyard pool

2 Copy and complete each conversion. PS R

a 7000 mL = _____ L
b 2.7 kL = _____ L
c 3.5 L = _____ mL
d 10 000 L = _____ kL
e 5200 kL = _____ ML
f 1.4 ML = _____ L
g 420 L = _____ kL
h 8.9 L = _____ mL
i 6.2 ML = _____ kL
j 1750 mL = _____ L
k 5 kL = _____ L
l 9000 kL = _____ ML
m 2500 kL = _____ L
n 520 L = _____ kL
o 23 mL = _____ L

3 What is the capacity of a regular bottle of cough medicine? Select the closest answer **A**, **B**, **C** or **D**.

A 200 mL **B** 500 mL **C** 1500 mL **D** 2000 mL

4 Match the correct capacity (**A** to **J**) with the items (**a** to **j**) listed. R

a car petrol tank
b a cup of flour
c bathtub
d bucket of water
e can of drink
f glass of water
g Olympic swimming pool
h bottle of soft drink
i teaspoon
j water storage tank

A 200 mL
B 23 kL
C 5 mL
D 70 L
E 1250 mL
F 1.875 ML
G 250 mL
H 9 L
I 375 mL
J 180 L

5 A jug holds 2 L of water. How many 250 mL glasses could be filled from it?

6 James is inviting 30 friends to a party. He calculates that each person will drink 1800 mL of soft drink. PS R

a How many litres of soft drink must he buy?
b If James intends to buy large 2 L bottles of drink, how many bottles must he buy?

7 Research the metric unit of capacity that is equal to 1000 megalitres. C

8 A lunch box is made in the shape of a rectangular prism. Its dimensions are 20 cm, 15 cm and 9 cm.

a Find the volume of the lunch box.

b How many mL of water would fit in the lunch box?

9 Gina's swimming pool is a rectangular prism 8 m long, 4 m wide and 1.5 m deep.

a Find the volume of the swimming pool.

b How many litres of water would be needed to fill the pool? (Hint: 1 m^3 holds 1 kL)

10 A fish tank is a rectangular prism 60 cm long, 40 cm high and 30 cm wide.

a Find the volume of the tank.

b How many litres of water will it hold?

11 A tap leaks 10 mL of water every 50 seconds. How much water will be lost in: PS R

a 1 second? **b** 1 minute? **c** 3 hours? **d** 1 day?

12 Leaving the tap on while brushing your teeth uses 5 litres of water. Assuming you brush your teeth twice a day, how many kilolitres of water are used in one year?
Select **A**, **B**, **C** or **D**. PS R

A 3.65 **B** 1825 **C** 3650 **D** 18 250

13 Your skin releases moisture as a way of controlling body temperature. On average, 200 mL is released per hour. If all this moisture was captured, how long would it take to fill a 1.25 L soft drink bottle? PS R

Did you know?

Water, water, everywhere

To help you better understand the size of a litre and a kilolitre, here are some examples of water use in and around the home:

• Washing your hands/face	5 L
• Brushing your teeth (tap running)	5 L
• Brushing your teeth (tap not running)	1 L
• Cooking and making coffee/tea	8 L per day
• Flushing the toilet	9 L to 13 L
• Flushing the toilet (half flush)	4.5 L to 6 L
• Household tap	18 L per minute
• Washing the dishes (hand)	18 L
• Washing the dishes (dishwasher)	25 L per cycle
• Bath	85 L to 150 L
• Shower (8 minutes)	80 L to 120 L
• Washing machine (front loading)	120 L per cycle

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• Washing machine (top loading)	180 L per cycle
• Washing the car (with hose)	100 L to 300 L
• Garden sprinkler	1 kL to 1.5 kL per hour
• Garden hose	1.8 kL per hour
• Swimming pool (backyard)	20 kL to 55 kL
• 50 m swimming pool	1870 kL

On average, a 4-person Sydney house (with garden) uses 936 litres of water per day. Half of it is used by outside taps or is flushed down a toilet.

How much water does your household use each day? Find out by asking your parents to show you the water bill.

Technology

Brogo and Windamere dams

In this activity, we will examine the volume of water in the Brogo (Bega) and Windamere (Mudgee) dams over the 12 months of a year.

	A	B	C	D	E	F	G	H	I
1	**Brogo Dam 2018**	Capacity=	8989	ML		**Windanere Dam 2018**	Capacity=	368120	ML
2									
3	Month	Volume (ML)	% Capacity			Month	Volume (ML)	% Capacity	
4	Jan	9002				Jan	166616		
5	Feb	8509				Feb	161735		
6	Mar	8964				Mar	158495		
7	Apr	8813				Apr	155636		
8	May	8068				May	152683		
9	Jun	7796				Jun	150841		
10	Jul	7818				Jul	150172		
11	Aug	7928				Aug	149070		
12	Sep	6091				Sep	148176		
13	Oct	5567				Oct	146595		
14	Nov	5538				Nov	144442		
15	Dec	6363				Dec	141623		
16									
17	*Source: www.water.nsw.gov.au*								
18									

1. Start a new spreadsheet and enter only the labels and values in rows 1 and 3. (Note: ML = megalitres.)
2. Enter **1 Jan 2018** in cell A4 and **1 Feb 2018** in cell A5. **Highlight** cells A4 and A5, right-click and **Fill Down** until you finish at **1 Dec 2018**.
3. Repeat (or copy) this for column F, cells F4 to F15.
4. Enter the volume amounts in columns B and G.
5. 'Capacity' refers to the total capacity of a dam. '% Capacity' calculates the volume as a percentage of the total capacity. To find the percentage capacity on **1 January 2018**, enter **=B4/C1** in cell C4 and click %.

6 **Fill Down** the formula from cell C4 to C15. Notice that, by using an absolute cell reference (C1), each volume is divided by the same number (the dam's total capacity).

C16

	A	B	C	D
1	**Brogo Dam 2018**	Capacity=	8989	ML
2				
3	Month	Volume (ML)	% Capacity	
4	Jan	9002	100%	
5	Feb	8509	95%	
6	Mar	8964	100%	
7	Apr	8813	98%	
8	May	8068	90%	

7 Repeat the steps above in cells H4 to H15 to calculate Windamere Dam's percentage capacity.

8 In cell A18, explain (in one sentence) why it is inappropriate to graph the % capacity for both dams on a single graph.

9 State the driest month for:

a Brogo Dam (answer in cell A19) **b** Windamere Dam (answer in B19).

10 State the wettest month for:

a Brogo Dam (A20) **b** Windamere Dam (B20).

11 Use the '% Capacity' column to find the month that shows the lowest dam level for each dam. Answer in cell A21 (Brogo) and B21 (Windamere).

12 Describe what happened at Brogo Dam from February to March. (A22)

13 What do the % capacity values from March to December show? (A23)

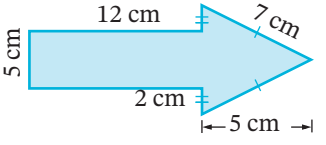
14 Do you think this data can be used to find out which area is more suitable for the development of a new town in the region? Begin your answer in cell A24. Give reasons to support your answer.

15 Which dam seems to be located in an area more likely to receive rain? Give reasons to support your answer. Answer in cell A25.

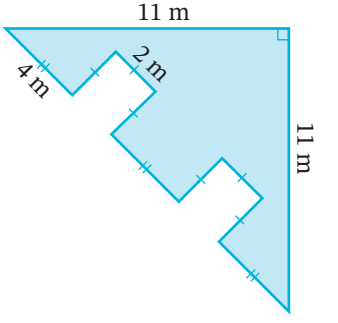
Power plus ANSWERS ON P. 590

1 Find the perimeter and area of each shape.

a

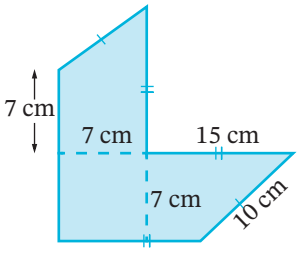


b

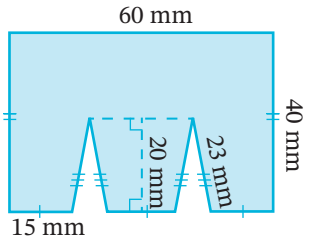


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c



d

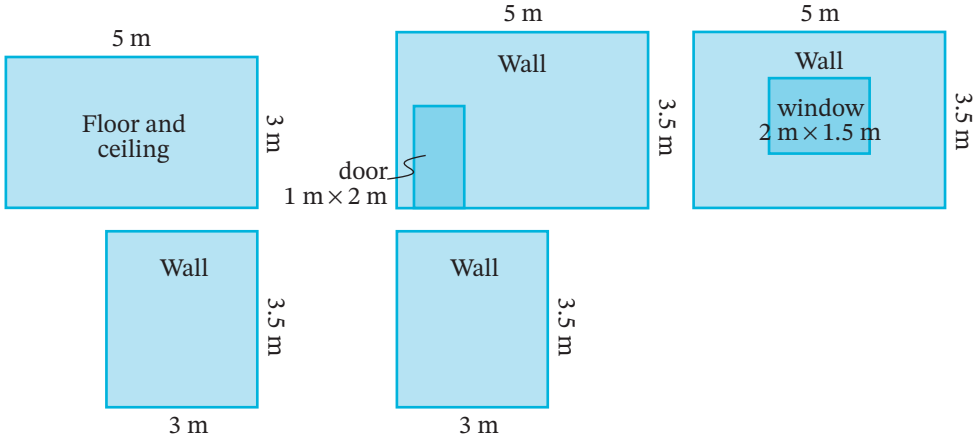


2 A rectangle has a perimeter of 30 cm.

- **a** List some possible dimensions for this rectangle.
- **b** For each pair of dimensions, calculate the area of the rectangle.
- **c** Which dimensions give the greatest area?

3 Find the length and width of a rectangle whose perimeter value is equal to its area value.

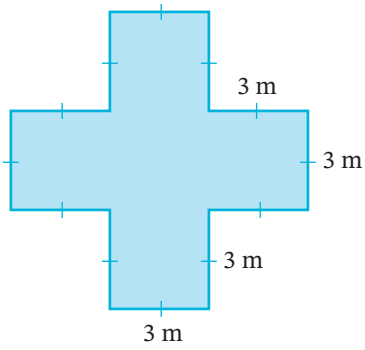
4 A room has dimensions as shown.



- **a** The 4 walls and the ceiling are to be painted (not including the door and window). Calculate how many square metres are to be painted.
- **b** Each surface requires 2 coats of paint, and 1 L of paint covers approximately 12 m^2. How many litres of paint are needed?
- **c** Paint comes in 4 L cans that cost $45.95. Calculate how many 4 L cans are needed and calculate the cost of the paint.
- **d** The floor is to be carpeted. How many square metres of carpet are needed?

5 A children's pool is in the shape of a cross, as shown on the right. Each side is 3 m long. The pool is filled with water to a depth of 300 mm.

- **a** Find the area of the pool's surface.
- **b** Calculate the volume of water in cubic metres (m^3).
- **c** If water is charged at $0.80 per kL, how much does it cost to fill the pool?



CHAPTER 8 REVIEW

Language of maths

Quiz
Language of maths 8

Puzzle
Area and volume crossword

area	breadth	capacity	centi-
circumference	cross-section	cubic	diameter
isometric	kilo-	length	mega-
micro-	milli-	parallelogram	perimeter
perpendicular height	pi (π)	prefix	prism
radius	square	volume	width

1 Which prefix in metric units means:

a one thousandth? **b** one thousand?

2 In your own words, explain how you would calculate the perimeter of a rectangle.

3 Why are there 2 formulas for the circumference of a circle?

4 What does **perpendicular height** mean?

5 Complete: A prism has a uniform _____.

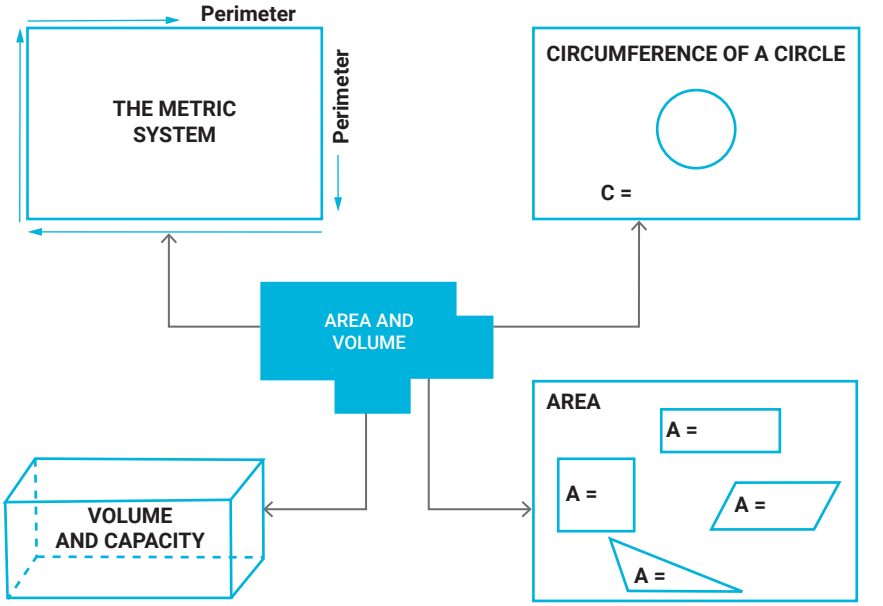
6 What is the difference between volume and capacity?

Topic summary

Worksheet
Mind map: Area and volume

- What have you learnt about measurement?
- What types of people use area as part of their jobs?
- List 2 measuring situations in which accuracy is important.
- Is there anything that you did not understand about the topic? Ask a friend or your teacher for help.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



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TEST YOURSELF 8

ANSWERS ON P. 590

Quiz
Test yourself 8

1 Convert: 8.01

a 240 min to h **b** 0.042 kg to g **c** 125 L to kL

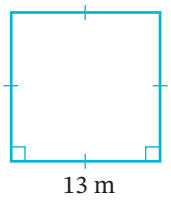
d 3200 m to km **e** 8.3 L to mL **f** 904 cm to mm

g 3.5 kg to t **h** 15 h to min **i** 15 h to s

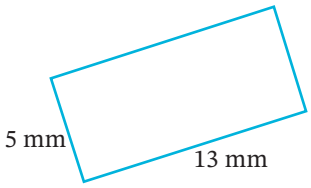
j 7150 mm to m **k** 4.2 kL to mL **l** 17 000 000 mg to kg

2 Find the perimeter of each shape. 8.02

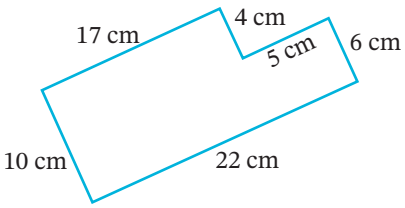
a



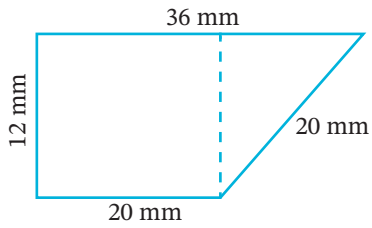
b



c

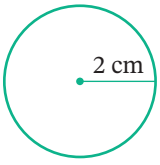


d

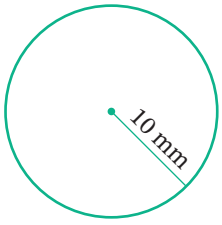


3 Calculate, correct to one decimal place, the circumference of each circle. 8.03

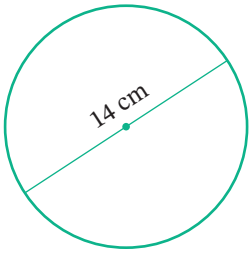
a



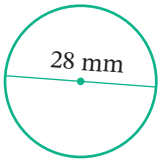
b



c



d



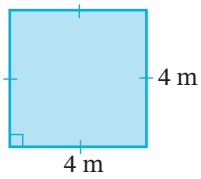
4 How many: 8.04

a cm^2 in 9.1 m^2? **b** mm^2 in 2.5 cm^2? **c** m^2 in 175 000 cm^2?

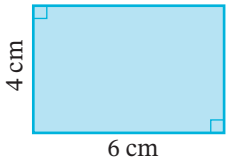
d m^2 in 240 000 mm^2? **e** cm^2 in 240 mm^2? **f** mm^2 in 8.6 m^2?

5 Find the area of each shape. 8.05

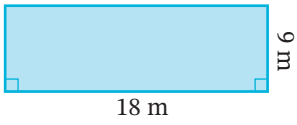
a



b



c



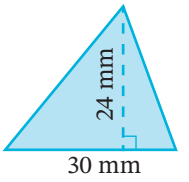
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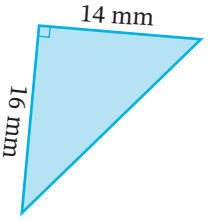
8.06

6 Find the area of each shape.

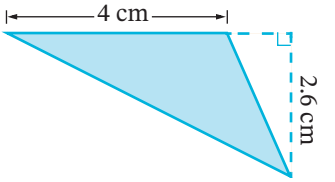
a



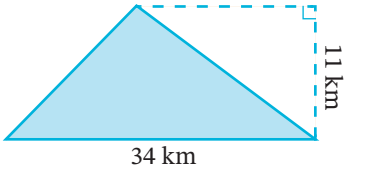
b



c



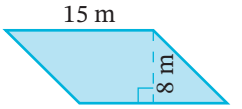
d



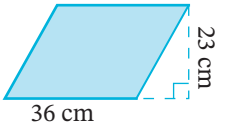
8.07

7 Find the area of each parallelogram.

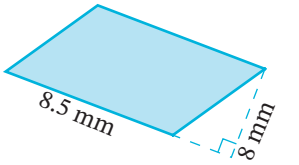
a



b



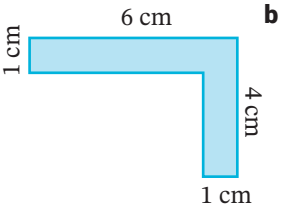
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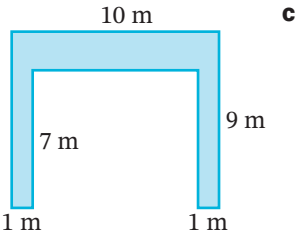
8.08

8 Find the shaded area in each shape.

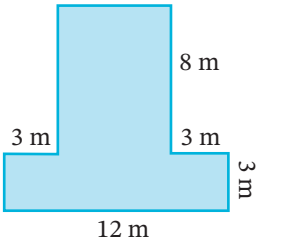
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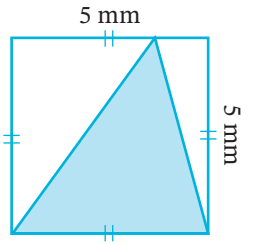
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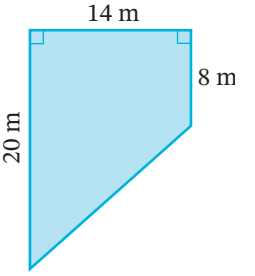
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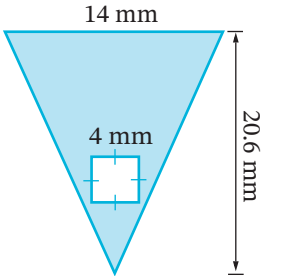
d



e

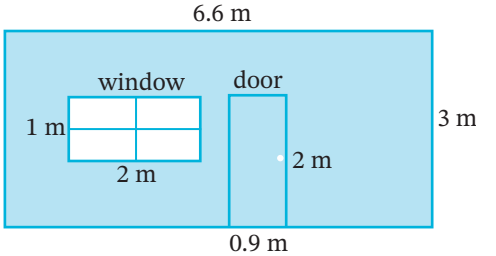


f



8.08

9 The wall shown needs painting. If one litre of paint covers 3 m^2 of wall, how many whole litres should be bought?



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10 Draw a cross-section of each solid and state whether it is a prism. 8.09

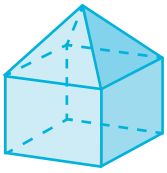
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b



c



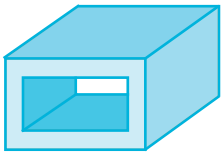
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e

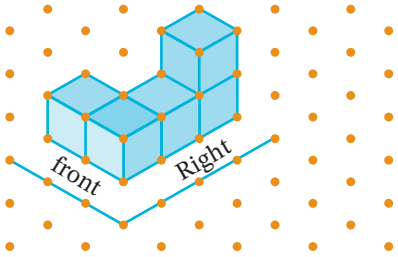


f



11 For each solid, sketch each view requested. 8.09

a

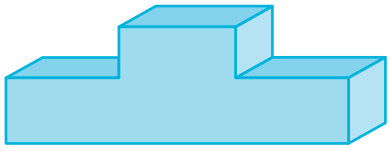


i front view

ii left view

iii top view

b



i right view

ii front view

iii top view

12 Copy and complete each conversion. 8.10

a $20\ cm^3 =$ _____ mm^3

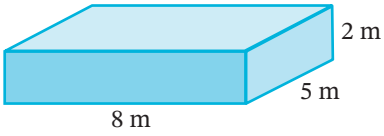
b $0.5\ m^3 =$ _____ cm^3

c $7500\ mm^3 =$ _____ cm^3

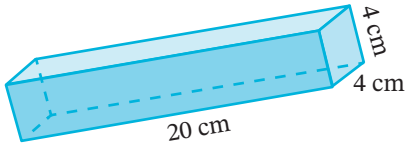
d $230\ 000\ mm^3 =$ _____ m^3

13 Find the volume of each prism. 8.11

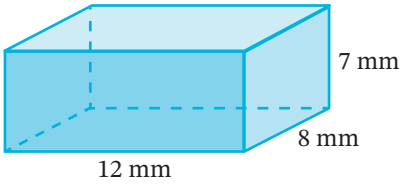
a



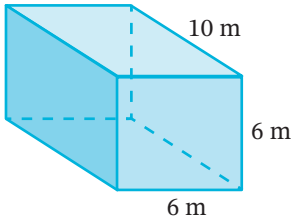
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c



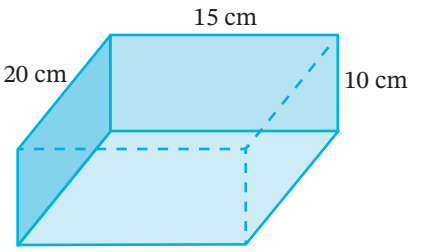
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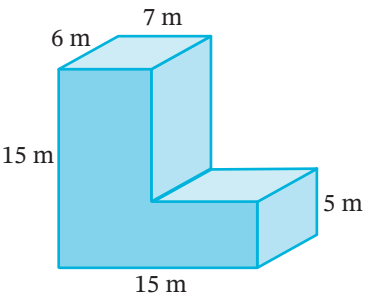
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e



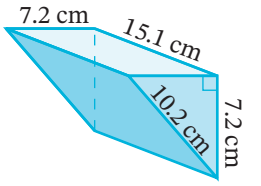
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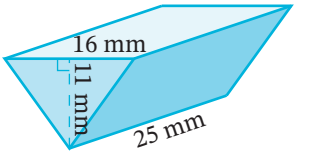
8.13 **14** Find the capacity, in litres, of each prism in question **13**.

8.12 **15** Find the volume of each prism.

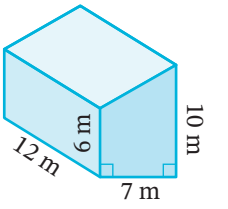
a



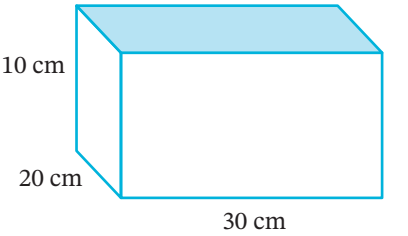
b



c



8.13 **16** This rectangular prism was filled with water.

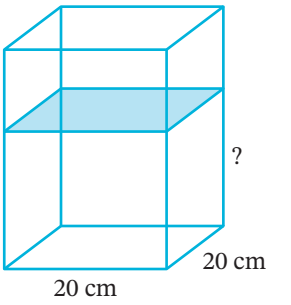


a Find the capacity of the prism in litres.

b The water was then poured into another rectangular prism with base 20 cm by 20 cm. To what height will the water reach? Select the correct answer **A**, **B**, **C** or **D**.

A 0.02 cm **B** 2400 cm

C 150 cm **D** 15 cm



8.01 **17** How many minutes has Isla lived by her 12th birthday? Assume all years have 365 days. Select **A**, **B**, **C** or **D**.

A 525 600 **B** 6 307 200 **C** 5 522 880 **D** 378 432 000

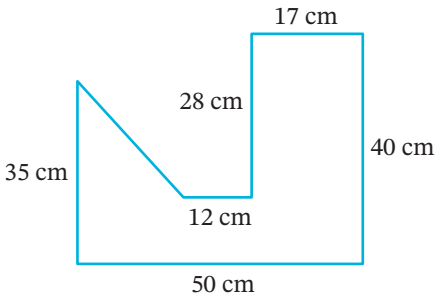
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18 Find the area of this shape. Select **A**, **B**, **C** or **D**. 8.08

A 1034 cm^2 **B** 1047.5 cm^2

C 1317.5 cm^2 **D** 1521.5 cm^2



19 The biggest iceberg on record was called B9. It had the same volume as a rectangular prism with dimensions 160 km long, 50 km wide and 250 metres high. When B9 melted, how many megalitres of water were contained in B9? 8.13



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